

Lecture# 23

MAXIMUM AND MINIMUM VALUES OF FUNCTIONS

- Absolute Extrema
- Finding Absolute Extrema for a continuous function
- Summary of extreme behaviors of functions over (a,b)
- Applied maximum and minimum problems
- Problems involving finite closed intervals
- Problems involving intervals that are non-finite and closed

1) ABSOLUTE EXTREMA

Previously we talked about relative maxima, relative minima of functions. These were like the highest mountain and the deepest valley in a given vicinity or neighborhood. Now we will talk about absolute maximum and minimum values of functions. These are like the highest peak in a mountain range, and the deepest valley.

Absolute maximum means that the value is the maximum one over the entire domain of the function.

Absolute minimum means that the value is the minimum one over the entire domain of the function.

If we think of the Earth's surface as defining some function, then its absolute maximum will be Mt. Everest, and absolute minimum will be the Marianna Trench in the Pacific Ocean near Hawaii.

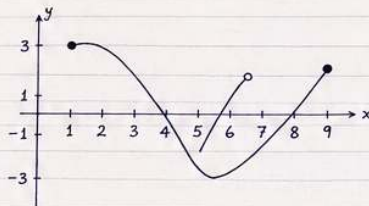
The question of interest given a function $f(x)$ is:

- 1) Does $f(x)$ has a maximum (minimum) value?
- 2) If $f(x)$ has a maximum (minimum), what is it?
- 3) If $f(x)$ has a maximum value, where does it occur?

Won't prove this as its difficult. Just use it. This theorem doesn't tell us what the max and the min are, just the conditions on a function which will make it have a max or min.

Example

Here is another function graphed over the interval $[1, 9]$. Although the interval is closed, the function not continuous on this interval as we can see from the graph, and so has no maximum or minimum values on that interval.



How to find Absolute Max and Min on a closed interval $[a,b]$

- Step 1: Find the critical points of f in (a,b)
- Step 2: Evaluate f at all the critical points and the endpoint a and b
- Step 3: The largest of the values in Step 2 is the maximum value of f on $[a,b]$ and the smallest is the minimum.

Example

Find the maximum and the minimum values of $f(x) = -x^3 + 2x^2 + 15$ on the interval $[1, 5]$

Solution:

- Since f is a polynomial, it's continuous and differentiable on the interval $(1, 5)$.

1) Find critical points:

$$f'(x) = -3x^2 + 4x$$

$$f'(x) = -3x^2 + 4x = 0$$

$$x(-3x + 4) = 0$$

$$x = 0, x = \frac{4}{3}$$

Only $x = \frac{4}{3}$ lies in $(1, 5)$. So critical point in $(1, 5)$ is $x = \frac{4}{3}$.

2) Evaluate f at critical point and endpoints $x = 1, x = \frac{4}{3}, x = 5$

$$f(1) = -(1)^3 + 2(1)^2 + 15 = -1 + 2 + 15 = 16$$

$$f\left(\frac{4}{3}\right) = -\left(\frac{4}{3}\right)^3 + 2\left(\frac{4}{3}\right)^2 + 15$$

$$= -\frac{64}{27} + 2 \cdot \frac{16}{9} + 15$$

$$= -\frac{64}{27} + \frac{32}{9} + 15$$

$$= -\frac{64}{27} + \frac{96}{27} + \frac{405}{27}$$

$$= \frac{437}{27} \approx 16.19$$

$$f(5) = -(5)^3 + 2(5)^2 + 15$$

$$= -125 + 50 + 15 = -60$$

3) Compare the values:

$$f(1) = 16$$

$$f\left(\frac{4}{3}\right) \approx 16.19$$

$$f(5) = -60$$

$$\text{Maximum value} = \frac{437}{27} \approx 16.19$$

$$\text{occurs at } x = \frac{4}{3}$$

$$\text{Minimum value} = -60$$

$$\text{occurs at } x = 5$$

SUMMARY OF EXTREME BEHAVIORS OF FUNCTIONS OVER (a,b)

Example

Find the max and min values if any, of the function $f(x) = x^4 + 2x^3 - 1$ on the interval $(-\infty, +\infty)$

Solution:

This is a continuous function on the given interval and

$$\lim_{x \rightarrow +\infty} (x^4 + 2x^3 - 1) = +\infty \quad \lim_{x \rightarrow -\infty} (x^4 + 2x^3 - 1) = +\infty$$

So f has a minimum but no maximum on $(-\infty, +\infty)$.

By Theorem 4.6.5, the min must occur at a critical point.

$$f'(x) = 4x^3 + 6x^2$$

$$f'(x) = 2x^2(2x + 3)$$

$$f'(x) = 0 \Rightarrow 2x^2(2x + 3) = 0$$

$$2x^2 = 0 \Rightarrow x = 0$$

$$2x + 3 = 0 \Rightarrow x = -\frac{3}{2}$$

Critical points: $x = 0, x = -\frac{3}{2}$

Evaluate f at critical points:

$$f(0) = 0^4 + 2(0)^3 - 1 = -1$$

$$f\left(-\frac{3}{2}\right) = \left(-\frac{3}{2}\right)^4 + 2\left(-\frac{3}{2}\right)^3 - 1$$

$$= \frac{81}{16} + 2\left(-\frac{27}{8}\right) - 1$$

$$= \frac{81}{16} - \frac{54}{8} - 1$$

$$= \frac{81}{16} - \frac{108}{16} - \frac{16}{16}$$

$$= -\frac{43}{16}$$

$$\text{Minimum value} = -\frac{43}{16}$$

at $x = -\frac{3}{2}$
No Maximum

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Lecture # 24

Newton's Method, Rolle's Theorem and Mean Value Theorem

- * Newton's method for approximating solutions to $f(x)=0$
- * Some difficulties with Newton's method
- * Rolle's Theorem
- * Mean Value Theorem

1. Newton's method for approximating solutions to $f(x)=0$

We have seen in algebra that the solution for the equation $ax+b=0$ is $x=-\frac{b}{a}$.

Similarly we have algebraic formulas for polynomial equations up to degree 5. But there is no algebraic solution to equations of the kind

$$x - \cos x = 0$$

For this equation and many like it, we settle for approximate solutions. How do we approximate the solutions? There are many methods, and one of them is Newton's method. Here is how Newton's method works to find approximate solutions to equations.

What does it mean for an equation to have a solution or that $f(x)=0$?

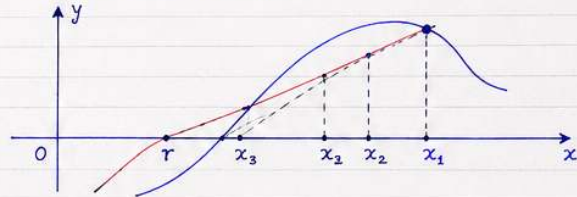
It means that we are looking for those x values, for which the corresponding y value or $f(x)$ is 0. This means that the solutions are those points where the graph of the function crosses the x -axis. Suppose that $x=r$ is the solution we are looking for. Let's approximate it by an initial guess called x_1 .

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We draw a line tangent to the graph of the given function at the point $(x_1, f(x_1))$.

If the tangent line is not parallel to the x -axis, then it will eventually intersect the x -axis at some point x_2 which will generally be closer to r than x_1 . We will repeat the process, with a tangent line at x_2 that meets x -axis at x_3 , and so on...

This is Newton's method.



We need a formula for this method.

Note that the tangent line at x_1 has the equation

$$y - f(x_1) = f'(x_1)(x - x_1) \quad (*)$$

If $f'(x_1) \neq 0$, then the line meets the x -axis at $(x_2, 0)$.

Plug this coordinate into above equation, we get

$$0 - f(x_1) = f'(x_1)(x_2 - x_1)$$

$$\Rightarrow x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \quad (1)$$

Repeating this process for a third point $(x_3, 0)$ gives

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} \quad (2)$$

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In general, then we have

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

There is limiting process involved here for finding the solutions. We get as close as we like to the solution.

* Example

The equation $x = \cos x$ has a solution between 0 and 1. Approximate it using Newton's method.

Solution:

Rewrite as $x - \cos x = 0$ so our function is $f(x) = x - \cos x$.

The derivative is $f'(x) = 1 + \sin x$.

So we have

$$x_{n+1} = x_n - \frac{\cos x_n}{1 + \sin x_n} \quad (\text{Formula})$$

Here is a graph of the situation. From the graph it looks like the solution is closer to 1 than 0. So we will use $x_1 = 1$.

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So we get the following approximations.

1st approximation ($n=1$):

$$\begin{aligned} x_2 &= x_1 - \frac{\cos x_1}{1 + \sin x_1} \\ &= 1 - \frac{\cos(1)}{1 + \sin(1)} \\ &= 1 - \frac{0.5403}{1 + 0.8415} \\ &= 1 - \frac{0.5403}{1.8415} \\ &= 1 - 0.2936 \\ x_2 &\approx 0.7503 \end{aligned}$$

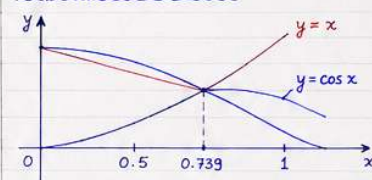
2nd approximation ($n=2$):

$$\begin{aligned} x_3 &= x_2 - \frac{\cos x_2}{1 + \sin x_2} \\ &= 0.7503 - \frac{\cos(0.7503)}{1 + \sin(0.7503)} \\ &= 0.7503 - \frac{0.7317}{1 + 0.6816} \\ &= 0.7503 - \frac{0.7317}{1.6816} \\ &= 0.7503 - 0.4356 \\ x_3 &\approx 0.7391 \end{aligned}$$

You may continue if you will, but we will say that the solution is approximately $x \approx 0.7391$

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Graph of the situation:



The curves intersect at $x \approx 0.7391$, which is the solution of $x = \cos x$.

Some difficulties with Newton's method

- Newton's method does not always work.
- If for some values of n , $f'(x_n) = 0$ then the formula for Newton's method involves division by 0 and we are out of business.
- Such a case will occur if the tangent line for some approximation has slope 0 or is parallel to the x -axis.
- Sometimes the approximations don't converge to a solution.

(To be continued in Part - 2)

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2. Rolle's Theorem

* Statement:

If a function $f(x)$ satisfies the following conditions:

- (i) $f(x)$ is continuous on $[a, b]$
- (ii) $f(x)$ is differentiable on (a, b)
- (iii) $f(a) = f(b)$

then there exists at least one point c in (a, b) such that

$$f'(c) = 0$$

In simple words:

If a smooth curve starts and ends at the same height, then somewhere in between it must have a horizontal tangent (slope = 0).

Example:

Let $f(x) = \sin x$ on $[0, 2\pi]$

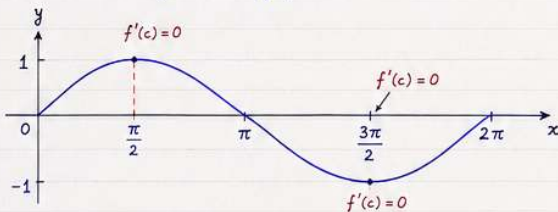
- (i) $\sin x$ is continuous on $[0, 2\pi]$ ✓
- (ii) $\sin x$ is differentiable on $(0, 2\pi)$ ✓
- (iii) $f(0) = \sin 0 = 0$ and $f(2\pi) = \sin 2\pi = 0$

So, by Rolle's Theorem, $\exists c \in (0, 2\pi)$ such that $f'(c) = 0$

Now, $f'(x) = \cos x$

$$\cos c = 0 \Rightarrow c = \frac{\pi}{2}, \frac{3\pi}{2} \quad (\text{both are in } (0, 2\pi))$$

Which is clear from the graph:



Another way to think about Rolle's theorem:

I leave Lahore and start driving towards Islamabad. Total distance between Lahore and Islamabad = 376 km.

- When I start driving, my velocity is 0.
- When I reach Islamabad, my velocity is again 0.
- Velocity is a continuous function on $[0, 376]$.
- Velocity is differentiable on $(0, 376)$ (since acceleration exists at every point).

By Rolle's Theorem, there is at least one point on my journey where acceleration = 0. (i.e. slope of velocity-time graph is 0)

Key Points (Rolle's Theorem)

- * Function must be continuous on $[a, b]$
- * Function must be differentiable on (a, b)
- * Function must have same value at a and b
- * Then $\exists c \in (a, b)$ such that $f'(c) = 0$

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3. Mean Value Theorem (MVT)

Statement:

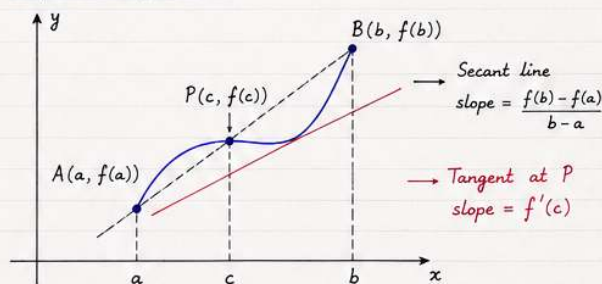
If $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) , then there exists at least one point $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

In simple words:

Under the right conditions, the slope of the tangent line at some point inside (a, b) is equal to the slope of the secant line joining the points $(a, f(a))$ and $(b, f(b))$.

Graphical Explanation:



Proof:

Define a new function

$$v(x) = f(x) - \left[\frac{f(b) - f(a)}{b - a} \right] (x - a) - f(a) \quad \text{for } x \in [a, b]$$

Step 1: $v(x)$ is continuous on $[a, b]$

(Since $f(x)$ is continuous on $[a, b]$)

Step 2: $v(x)$ is differentiable on (a, b)

(Since $f(x)$ is differentiable on (a, b))

Step 3: $v(a) = 0$ and $v(b) = 0$

$$v(a) = f(a) - \left[\frac{f(b) - f(a)}{b - a} \right] (a - a) - f(a) = f(a) - 0 - f(a) = 0$$

$$v(b) = f(b) - \left[\frac{f(b) - f(a)}{b - a} \right] (b - a) - f(a) = f(b) - (f(b) - f(a)) - f(a) = f(b) - f(b) + f(a) - f(a) = 0$$

Step 4: Apply Rolle's Theorem to $v(x)$ on $[a, b]$

Since $v(a) = v(b) = 0$ and $v(x)$ satisfies the conditions of Rolle's Theorem, $\exists c \in (a, b)$ such that $v'(c) = 0$

Step 5: Differentiate $v(x)$

$$v'(x) = f'(x) - \left[\frac{f(b) - f(a)}{b - a} \right]$$

Step 6: Put $x = c$

$$v'(c) = 0 \Rightarrow f'(c) - \left[\frac{f(b) - f(a)}{b - a} \right] = 0$$

$$\Rightarrow f'(c) = \frac{f(b) - f(a)}{b - a}$$

Q.E.D.

Key Points (MVT)

- * $f(x)$ must be continuous on $[a, b]$
- * $f(x)$ must be differentiable on (a, b)
- * Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Geometrically:

Slope of tangent at some point = slope of secant line.

This theorem is very powerful and has many applications in Calculus, Physics, Economics, and Engineering. 😊

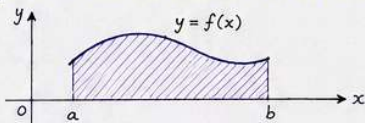
Lecture # 25 INTEGRATION

In this lecture we will look at the beginnings of the other major Calculus problem.

- The Area Problem
- Anti-derivatives (Integration)
- Integration formulas
- Indefinite Integral
- Properties of Indefinite Integral

1) THE AREA PROBLEM

Given a continuous and non-negative function $f(x)$ on an interval $[a, b]$, find the area between the graph of f and the interval $[a, b]$ on the x -axis.



Instead of trying to solve a particular case like the one in the picture, we will generalize to solve this problem where the right end point will be any number $x \geq a$ instead of just b . We denote the area we are trying to find as $A(x)$ because this will be a function of x as it depends on how far away x is from a . It was the idea of Newton and Leibniz that to find the unknown area $A(x)$, first find its derivative $A'(x)$ and use this derivative to determine what $A(x)$ is!

So we want to find out first.

$$A'(x) = \lim_{h \rightarrow 0} \frac{A(x+h) - A(x)}{h}$$

Let's assume for now that $h > 0$

The top of the derivative quotient is the difference of the two areas $A(x)$ and $A(x+h)$.

Let c be the midpoint between x and $x+h$.

Then the difference of areas can be approximated by the area of the rectangle with base length h and height $f(c)$.

So we have

$$A(x+h) - A(x) \approx f(c) \cdot h \Rightarrow \frac{A(x+h) - A(x)}{h} \approx f(c)$$

Note that the error in the approximation from this rectangle in A will approach 0 as $h \rightarrow 0$.

Then we have

$$A'(x) = \lim_{h \rightarrow 0} \frac{A(x+h) - A(x)}{h} = \lim_{h \rightarrow 0} f(c) = f(x)$$

As $h \rightarrow 0$, c approaches x . Also, f is continuous, so $f(c) \rightarrow f(x)$ as $c \rightarrow x$. Thus

$$\therefore A'(x) = f(x)$$

So: The derivative of the area function $A(x)$ is the function whose graph forms the upper boundary of the region under which the area is to be found.

Example 1

Find the area of the region under the graph of $y = f(x) = x^2$ over the interval $[0, 1]$.

Look at the situation over the interval $[0, x]$.

Then from the discussion, we have $A'(x) = x^2$

To find $A(x)$ we look for a function whose derivative is x^2 . This is called an antidifferentiation problem.

A guess is the function $A(x) = \frac{x^3}{3}$

$$\text{Check: } A'(x) = \left(\frac{x^3}{3}\right)' = \frac{3x^2}{3} = x^2 \quad \checkmark$$

Since $A(0) = 0$, the constant is 0.

So, $A(x) = \frac{x^3}{3}$ is the area function.

On the interval $[0, 1]$, put $x = 1$:

$$A(1) = \frac{1^3}{3} = \frac{1}{3} \text{ square units}$$

2) ANTI-DERIVATIVES

Definition 5.2.1

A function F is called antiderivative of a function f on a given interval if $F'(x) = f(x)$ for all x in the interval.

Example 2

The functions $\frac{x^3}{3} - 1$, $\frac{x^3}{3} + 1$, $\frac{x^3}{3} + \pi$

are all anti-derivatives of $f(x) = x^2$ on the interval $(-\infty, \infty)$

Because the derivative of each is:

$$\left(\frac{x^3}{3} - 1\right)' = \left(\frac{x^3}{3} + 1\right)' = \left(\frac{x^3}{3} + \pi\right)' = x^2 = f(x)$$

If $F(x)$ is any anti-derivative of $f(x)$, then so is $F(x) + C$, where C is a constant.

3) INDEFINITE INTEGRAL

Theorem:

If there is some function F such that $[F(x)]' = f(x)$, then all functions of the form $F(x) + C$ are anti-derivatives of $f(x)$.

We denote this by $\int f(x) dx = F(x) + C$

The symbol $\int$ is called the integral sign and $f(x)$ is called the integrand.

It is read as: "Indefinite integral of $f(x)$ equals $F(x)$ "

The right side is not a specific function but a whole set of possible functions. That's why we call it the indefinite integral.

C is called the constant of integration.

Example 3

The anti-derivatives of $f(x) = x^2$ are functions of the form

$$F(x) = \frac{x^3}{3} + C$$

So we can write $\int x^2 dx = \frac{x^3}{3} + C$

The dx identifies the independent variable in the integration.

4) INTEGRATION FORMULAS (From Table)

$$\begin{aligned} \bullet \int x^3 dx &= \frac{x^4}{4} + C & \bullet \int \frac{1}{x} dx &= \ln|x| + C \\ \bullet \int x^4 dx &= \frac{x^5}{5} + C & \bullet \int dx &= x + C \end{aligned}$$

Examples using formulas

$$\text{i) } \int x^3 dx = \frac{x^4}{4} + C \quad \text{iii) } \int \frac{1}{x} dx = \ln|x| + C$$

$$\text{ii) } \int x^4 dx = \frac{x^5}{5} + C \quad \text{iv) } \int dx = x + C$$

5) PROPERTIES OF INDEFINITE INTEGRAL

$$\int f(x) dx = F(x) + C \Rightarrow \frac{d}{dx} \left[\int f(x) dx \right] = f(x)$$

(Differentiation undoes Integration)

Examples

Example 4

Evaluate $\int 4 \cos x dx$

$$= 4 \int \cos x dx$$

$$= 4 [\sin x] + C$$

$$= 4 \sin x + C$$

$$(\text{Let } 4C = K)$$

$$= 4 \sin x + K$$

Example 5

Evaluate $\int (x^2 + x) dx$

$$= \int x^2 dx + \int x dx$$

$$= \frac{x^3}{3} + \frac{x^2}{2} + C$$

$$= \frac{x^3}{3} + \frac{x^2}{2} + C$$

6) GENERALIZED VERSION (Theorem 5.2.3 b and c)

$$\begin{aligned} \int [c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x)] dx \\ = c_1 \int f_1(x) dx + c_2 \int f_2(x) dx + \dots + c_n \int f_n(x) dx \end{aligned}$$

Example 6

Evaluate $\int (3x^6 - 2x^2 + 7x - 1) dx$

$$= \int 3x^6 dx - \int 2x^2 dx + \int 7x dx - \int 1 dx$$

$$= 3 \frac{x^7}{7} - 2 \frac{x^3}{3} + 7 \frac{x^2}{2} - x + C$$

$$= \frac{3x^7}{7} - \frac{2x^3}{3} + \frac{7x^2}{2} - x + C$$

Example 7

$$\text{a) } \int \cos x dx = \sin x + C$$

$$\text{b) } \int \sin x dx = -\cos x + C$$

$$\text{c) } \int e^x dx = e^x + C$$

$$\begin{aligned} \text{d) } \int \frac{\cos x}{\sin^2 x} dx \\ = \int \cot x (\csc x) dx \\ = -\csc x + C \end{aligned}$$

Lecture #26 INTEGRATION BY SUBSTITUTION

This is like the Chain Rule we saw for differentiation. The idea is to integrate functions that are composition of different functions.

$$\int f(u) du = G(u) \quad \text{where } G'(u) = f(u)$$

Also this means that

$$\int f(u) du = G(u) + C$$

Let u be a function of x . Then we have

$$\frac{d}{dx} [G(u)] = G'(u) \cdot \frac{du}{dx} = f(u) \frac{du}{dx}$$

Now if we apply the integral on both sides w.r.t. x we get

$$\int \frac{d}{dx} [G(u)] dx = \int f(u) \frac{du}{dx} dx$$

$$\Rightarrow G(u) = \int f(u) du = G(u) + C$$

$$\therefore \int f(u) du = G(u) + C$$

Example 1 $\int 2x(x^2+1)^{50} dx$

Let $u = x^2+1$. Then $\frac{du}{dx} = 2x \Rightarrow du = 2x dx$

Now we can rewrite the given problem as

$$\int 2x(x^2+1)^{50} dx = \int u^{50} (2x dx) = \int u^{50} du$$

We know $\int u^n du = \frac{u^{n+1}}{n+1} + C$

$$\text{So, } \int u^{50} du = \frac{u^{51}}{51} + C$$

Replace u by x^2+1 :

$$\int 2x(x^2+1)^{50} dx = \frac{(x^2+1)^{51}}{51} + C$$

Caution: Don't feel tempted to just add 1 to the power 50 in the original problem!! That will be incorrect. The reason is that the "Power Rule" for the integrals is applicable to function which are not a composition of others. In this example, $2x(x^2+1)^{50}$ is a composition of two functions.

GENERAL PROCEDURE (u-SUBSTITUTION)

1. Make the choice for u , say $u = g(x)$
2. Compute $\frac{du}{dx} = g'(x)$
3. Make the substitution $u = g(x)$, $du = g'(x) dx$ in the original integral.
By this point, the whole integral should be in terms of u and there should be no x 's in it.
4. Evaluate the resulting integral.
5. Replace u by $g(x)$ so the final answer is in x .

- No hard and fast rule for choosing u .
- The choice of u should be such that the resulting calculus results in a simplified integral.
- There may be more than one choice of u in some cases. Only practice and experience makes things easier.

Think of this choice of u like playing chess. You need to choose it so that the future looks bright for problem solution!

★ If the integrand is the derivative of a known function with a constant added or subtracted from the independent variable, then the substitution is the easiest.

Here is Table 5.2.1 (Basic Integration Formulas)

Function $f(x)$	$\int f(x) dx$	Function $f(x)$	$\int f(x) dx$
x^n ($n \neq -1$)	$\frac{x^{n+1}}{n+1} + C$	$\cos x$	$\sin x + C$
$\frac{1}{x}$	$\ln x + C$	$\sec^2 x$	$\tan x + C$
e^x	$e^x + C$	$\frac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x + C$
$\sin x$	$-\cos x + C$	$\frac{1}{1+x^2}$	$\tan^{-1} x + C$

We want to integrate a function in which

"The integrand is the derivative of a known function with a constant added or subtracted from the independent variable"

Example 2

$$\int \sin(x+9) dx$$

Let $u = x+9 \Rightarrow du = dx$

$$\int \sin(x+9) dx = \int \sin u du$$

$$= -\cos u + C$$

$$= -\cos(x+9) + C$$

The integrand sine is the derivative of cosine function

Example 3

$$\int 24(x-8)^{23} dx$$

Let $u = x-8 \Rightarrow du = dx$

$$\int 24(x-8)^{23} dx = \int 24u^{23} du$$

$$= 24 \cdot \frac{u^{24}}{24} + C = u^{24} + C$$

$$= (x-8)^{24} + C$$

Integrand is the derivative of a known function and a constant multiplies the independent variable.

Example 4

Let $u = 5x \Rightarrow du = 5 dx \Rightarrow dx = \frac{du}{5}$

$$\int \cos(5x) dx = \int \cos u \cdot \frac{du}{5}$$

$$= \frac{1}{5} \int \cos u du = \frac{1}{5} \sin u + C$$

$$= \frac{1}{5} \sin(5x) + C$$

Example 5

$$\int \sin^2(x) \cos(x) dx$$

Let $u = \sin x \Rightarrow du = \cos x dx$

$$\int \sin^2(x) \cos(x) dx = \int u^2 du$$

$$= \frac{u^3}{3} + C = \frac{\sin^3 x}{3} + C$$

Example 6

$$\int \frac{\cos(\ln x)}{x} dx$$

Let $u = \ln x \Rightarrow du = \frac{1}{x} dx$

$$\int \frac{\cos(\ln x)}{x} dx = \int \cos u du$$

$$= \sin u + C = \sin(\ln x) + C$$

Complicated example: $\int t^4 (3-5t^5)^3 dt$

What should the substitution be?

Idea: get an expression $= u$ so that when we differentiate it, we get the remaining part $t^4 dt$.

$$\text{Let } u = 3-5t^5 \Rightarrow du = -25t^4 dt$$

$$\Rightarrow t^4 dt = -\frac{1}{25} du$$

$$\int t^4 (3-5t^5)^3 dt = \int u^3 \left(-\frac{1}{25} du\right)$$

$$= -\frac{1}{25} \int u^3 du = -\frac{1}{25} \left(\frac{u^4}{4}\right) + C$$

$$= -\frac{1}{100} u^4 + C$$

$$= -\frac{1}{100} (3-5t^5)^4 + C$$

Lecture # 27

SIGMA NOTATION

- Sigma notation is used to write lengthy sums in compact form.
- Sigma or Σ is an Upper case letter in Greek.
- This symbol is called Sigma or summation as it is used to represent lengthy sums.

EXAMPLE 1 (How sigma notation works)

Consider the sum $1^2 + 2^2 + 3^2 + 4^2 + 5^2$

Note that every term in this sum is the square of an integer from 1 to 5.

Let's assign these integers a variable k and keep in mind that this k can take on values from 1 to 5.

Then we can say that k^2 represent each of the elements in the sum.

So we can write this as

$$\sum_{k=1}^5 k^2 \quad \text{"Summation of } k^2 \text{ where } k \text{ goes from 1 to 5"}$$

EXAMPLE 2

$$(i) \sum_{k=1}^5 3k = 3(1) + 3(2) + 3(3) + 3(4) + 3(5) = 3 + 6 + 9 + 12 + 15 = \boxed{45}$$

$$(ii) \sum_{k=0}^8 k^2 = 0^2 + 1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 = 0 + 1 + 4 + 9 + 16 + 25 + 36 + 49 + 64 = \boxed{204}$$

$$(iii) \sum_{k=1}^6 (2k-1) = (2(1)-1) + (2(2)-1) + (2(3)-1) + (2(4)-1) + (2(5)-1) + (2(6)-1) = 1 + 3 + 5 + 7 + 9 + 11 = \boxed{36}$$

☆ The number on the top are called the upper limits of the summation and the numbers at the bottom are called lower limits of the summation.

☆ The letter k is called the index of the summation.

☆ It is not necessary that the letter k represents the index of summation. We could use i or j or m etc.

$$\text{EXAMPLE 3} \quad \sum_{j=1}^4 j, \quad \sum_{i=1}^4 i, \quad \sum_{m=1}^4 m$$

All denote the sum $1 + 2 + 3 + 4 = \boxed{10}$

☆ If the upper and lower limits are the same, then the summation reduces to just one term.

$$\sum_{k=3}^3 k^2 = 3^2 = 9$$

☆ If the expression to the right of the summation does not involve the index of the summation, then do the following.

$$\sum_{k=1}^5 6 = 6 + 6 + 6 + 6 + 6 = 6(5) = 30$$

EXAMPLE 4 (A sum can be written in more than one way)

The following summations all represent the sum of the first five positive integers $= 1 + 2 + 3 + 4 + 5 = 15$

$\sum_{k=1}^5 k$ $= 1 + 2 + 3 + 4 + 5$ $= \boxed{15}$	$\sum_{k=0}^4 (k+1)$ $= (0+1) + (1+1) + (2+1) + (3+1) + (4+1)$ $= 1 + 2 + 3 + 4 + 5 = \boxed{15}$	$\sum_{k=2}^6 (k-1)$ $= (2-1) + (3-1) + (4-1) + (5-1) + (6-1)$ $= 1 + 2 + 3 + 4 + 5 = \boxed{15}$
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LECTURE #27 SIGMA NOTATION (Part-2)

5) CHANGING THE INDEX OF THE SUMMATION

It is often necessary and useful to change a given Sigma notation for a sum to another sigma notation with different limits of summation.

EXAMPLE 5

Express $\sum_{k=3}^7 (5k-2)$ in sigma notation so that the lower limit is 0 rather than 3.

Solution:

Define a new summation index j by the following formula

$$j = k - 3 \Rightarrow k = j + 3$$

Then as k runs from 3 to 7, j runs from 0 to 4.

$$\begin{aligned} \text{So, } \sum_{k=3}^7 (5k-2) &= \sum_{j=0}^4 [5(j+3)-2] \\ &= \sum_{j=0}^4 (5j+15-2) \\ &= \sum_{j=0}^4 (5j+13) \end{aligned}$$

Check:

Original sum $\sum_{k=3}^7 (5k-2)$	New sum $\sum_{j=0}^4 (5j+13)$
$= 5(3)-2+5(4)-2+5(5)-2+5(6)-2+5(7)-2$	$= 5(0)+13+5(1)+13+5(2)+13$
$= 13+18+23+28+33 = 115$	$+ 5(3)+13+5(4)+13$
	$= 13+18+23+28+33 = 115$

You should check that the two actually represent the same sum by putting values into the index in both notations.

6) REPRESENTING A GENERAL SUM

Example 6

a) $a_1 + a_2 + a_3$ represent the general sum with three terms.

This can also be written as in sigma notation

$$\sum_{k=1}^3 a_k = a_1 + a_2 + a_3$$

b) General sum with n terms can be written as

$$\sum_{k=1}^n b_k = b_1 + b_2 + b_3 + \dots + b_n$$

7) PROPERTIES OF SIGMA NOTATION

Here are a few properties of sigma notation that will be helpful later on.

$$\star \sum_{k=m}^n c a_k = c \sum_{k=m}^n a_k, \quad c \text{ is any constant}$$

$$\star \sum_{k=m}^n (a_k + b_k) = \sum_{k=m}^n a_k + \sum_{k=m}^n b_k$$

$$\star \text{ If } m > n, \sum_{k=m}^n a_k = 0$$

$$\star \sum_{k=m}^n a_k = \sum_{k=p}^{p+(n-m)} a_k \quad (p \text{ is any integer})$$

$$\text{eg. } \sum_{k=1}^5 a_k = \sum_{k=4}^8 a_k$$

$$\star \sum_{k=m}^n a_k = \sum_{t=1}^{n-1} a_{t+1}$$

8) SOME IMPORTANT SUM FORMULAS (Table 5.4.1)

	Formula	Sigma Form	Closed Form
(i)	$1 + 2 + 3 + \dots + n$	$\sum_{k=1}^n k$	$\frac{n(n+1)}{2}$
(ii)	$1^2 + 2^2 + 3^2 + \dots + n^2$	$\sum_{k=1}^n k^2$	$\frac{n(n+1)(2n+1)}{6}$
(iii)	$1^3 + 2^3 + 3^3 + \dots + n^3$	$\sum_{k=1}^n k^3$	$\left[\frac{n(n+1)}{2} \right]^2$
(iv)	$a + a + a + \dots + a$ (n terms)	$\sum_{k=1}^n a$	na
(v)	$1 + 1 + 1 + \dots + 1$ (n terms)	$\sum_{k=1}^n 1$	n
(vi)	$r + r^2 + r^3 + \dots + r^n$	$\sum_{k=1}^n r^k$	$\frac{r(r^n-1)}{r-1}, r \neq 1$
(vii)	$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)}$	$\sum_{k=1}^n \frac{1}{k(k+1)}$	$\frac{n}{n+1}$
(viii)	$r^0 + r^1 + r^2 + \dots + r^n$	$\sum_{k=0}^n r^k$	$\frac{r^{n+1}-1}{r-1}, r \neq 1$

Example 7 (Using formula ii)

$$\begin{aligned} \text{Evaluate } \sum_{k=1}^{30} k^2 &= \frac{30(30+1)(2(30)+1)}{6} \\ &= \frac{30(31)(61)}{6} = \frac{30(31)(61)}{2 \cdot 3} \\ &= 15(31)(61) = 465(61) \\ &= 28365 \end{aligned}$$

9) THEOREM 5.4.2 (a) and (b)

In a formula involving summation like this one

$$\begin{aligned} \text{a) } \sum_{k=1}^n (2k+1) &= \underbrace{(2 \cdot 1 + 1)}_{1^{\text{st}} \text{ term}} + \underbrace{(2 \cdot 2 + 1)}_{2^{\text{nd}} \text{ term}} + \dots + \underbrace{(2n+1)}_{n^{\text{th}} \text{ term}} \\ &= 1 + 3 + 5 + \dots + (2n+1) \quad (\text{Open form}) \\ &= n(n+1) \quad (\text{Closed form}) \end{aligned}$$

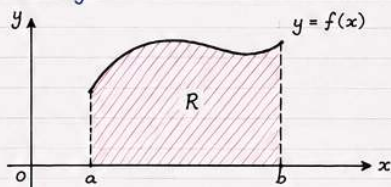
$$\begin{aligned} \text{b) } \sum_{k=1}^n (3k-2) &= (3 \cdot 1 - 2) + (3 \cdot 2 - 2) + \dots + (3n-2) \\ &= 1 + 4 + 7 + \dots + (3n-2) \quad (\text{Open form}) \\ &= \frac{n(3n-1)}{2} \quad (\text{Closed form}) \end{aligned}$$

Lecture # 28 AREA AS LIMITS

1) INTRODUCTION

- Definition of Area
- Some technical considerations
- Numerical approx of area

Look at the figure below.



In this figure, there is a region bounded below by the x-axis, on the sides by the lines $x=a$ and $x=b$, and above by a curve or the graph of a continuous function $y=f(x)$ which is also non-negative on the interval $[a, b]$.

Earlier we saw that such an area can be computed using anti-derivatives.

2) MAKING THE CONCEPT PRECISE

Recall that the slope of the tangent line was defined in terms of the limit of the slopes of secant lines.

Similarly, we will define area of a region R as limits of the areas of simpler regions whose areas are known. We will break up R into rectangles, and then find the area of each rectangle in R and then add all these areas up. The result will be an approximation to the region R . Let's call it R_n .

If we let n go to infinity, the resulting will give a better and better approximation to R as the rectangle in R will get thinner and thinner, and the gaps will be filled in.

Here is the formal idea.

Choose an arbitrary positive integer n , and divide the interval $[a, b]$ into n subintervals of width $\frac{b-a}{n}$ by inserting $n-1$ equally spaced points between a and b say $x_1, x_2, \dots, x_{n-1}$.

$$\Delta x = \frac{b-a}{n}$$

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These points of subdivision form a regular partition of $[a, b]$

$$a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b$$

Next draw a vertical line through the points $x_1, x_2, \dots, x_{n-1}$.

This will divide the region R into n strips of uniform width Δx .

Now we want to approximate the area of each strip by the area of a rectangle. For this, choose an arbitrary point x_k^* in each subinterval $[x_{k-1}, x_k]$.

Over each subinterval construct a rectangle whose height is the values of the function f at the selected arbitrary point.

The union of these rectangles form the region R_n that we can regard as a reasonable approx to the region R . The area of R_n can be got by adding the areas of all the rectangles forming it.

If now we let n get big, the number of rectangles gets big, and the gaps btw the curve and the rectangles get filled in.

As n goes to infinity, the approx gets as good as the real thing.

So we define

$$A(\text{area } R) = \lim_{n \rightarrow +\infty} [\text{area } R_n]$$

For all the following work and computation, we will treat n as a POSITIVE integer.

3) COMPUTATIONAL FORM OF AREA

In the interval $[a, b]$, each of the approximating rectangles has width $\frac{b-a}{n}$.

We call this Δx .

$$\Delta x = \frac{b-a}{n}$$

The heights of the approximating rectangles are at the points $x_1^*, x_2^*, \dots, x_n^*$.

So the approximating rectangles making up the region have areas

$$f(x_1^*)\Delta x, f(x_2^*)\Delta x, \dots, f(x_n^*)\Delta x$$

4) SUM OF AREAS AND DEFINING FORMULA

So the total area of R_n is given by

$$\begin{aligned} \text{area } R_n &= f(x_1^*)\Delta x + f(x_2^*)\Delta x + \dots + f(x_n^*)\Delta x \\ &= \Delta x [f(x_1^*) + f(x_2^*) + \dots + f(x_n^*)] \\ &= \Delta x \sum_{k=1}^n f(x_k^*) \end{aligned}$$

So now A can be written as

$$A = \lim_{n \rightarrow +\infty} \sum_{k=1}^n f(x_k^*)\Delta x$$

This we will take to be the PRECISE definition of the area of the region R .

5) SOME TECHNICAL CONSIDERATIONS

- ★ The points $x_1^*, x_2^*, \dots, x_n^*$ were chosen arbitrarily. What if some other points were chosen? Would the resulting values for $f(x)$ at these points be different? And if so then the definition of area will not be well-defined?
- ★ It is proved in advanced courses that since f is continuous it doesn't matter what points are taken in a subinterval.
- ★ Usually the point x_k^* in a subinterval is chosen so that it is the left end point, the right end point or the midpoint of the interval.
- ★ Note that we divided the interval $[a, b]$ by the points $x_1, x_2, \dots, x_{n-1}$ with $x_0 = a$ and $x_n = b$ into equal parts of width Δx .

So

- 1) Left end point $x_k^* = x_{k-1} = a + (k-1)\Delta x$
- 2) Right end point $x_k^* = x_k = a + k\Delta x$
- 3) Midpoint $x_k^* = x_{k-1} + \frac{\Delta x}{2} = a + \left(k - \frac{1}{2}\right)\Delta x$

EXAMPLE 1 (Right end points)

Use the definition of the Area with x_k^* as the right end point of each subinterval to find the area under the line $y = x$ over the interval $[1, 2]$.

Subdivide $[1, 2]$ into n equal parts, then each part will have length $\Delta x = \frac{2-1}{n} = \frac{1}{n}$.

We are taking x_k^* as the right end point so

$$x_k^* = a + k\Delta x = 1 + \frac{k}{n}$$

Thus, the k^{th} rectangle has area

$$f(x_k^*)\Delta x = x_k^* \cdot \Delta x = \left(1 + \frac{k}{n}\right)\left(\frac{1}{n}\right)$$

And the sum of the areas of the n rectangles is

$$\begin{aligned} A &= \lim_{n \rightarrow +\infty} \sum_{k=1}^n \left(1 + \frac{k}{n}\right)\left(\frac{1}{n}\right) \\ &= \lim_{n \rightarrow +\infty} \frac{1}{n} \left[\sum_{k=1}^n 1 + \frac{1}{n} \sum_{k=1}^n k \right] \end{aligned}$$

Lecture # 29 THE DEFINITE INTEGRAL (Part-1)

- Definition of Definite Integral
- Definite Integral of continuous functions with nonnegative values
- Definite Integral of continuous functions with negative and positive values
- Definite Integral of functions with discontinuities
- Properties of the Definite Integral
- Inequalities involving Definite Integral

1) INTRODUCTION

We have so far developed a definition of Area as limit of a bunch of rectangles with equal widths.

The question arises that why choose widths? The answer is that we don't have to.

We want now to have a definition of Area that includes the general case where the widths of the rectangles are not necessarily equal.

When the rectangles widths were equal note that the formula for the width was defined as

$$\Delta x = \frac{b-a}{n}$$

In this formula you can see that as n goes to infinity, the width Δx goes to zero.

If the widths are not the same for all rectangles, then this is not necessarily the case.

Suppose that we have a rectangle construction in which we divide $[a, b]$ so that one half is continually subdivided into smaller rectangle, while the right is left as one big rectangle.

In this case, when n goes to ∞ , left part goes to 0, not the Right half.

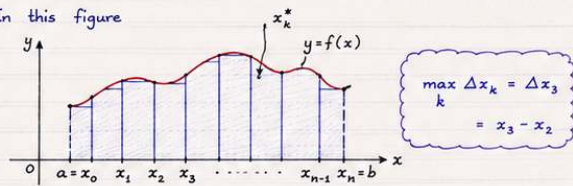
Let's fix this problem.

Subdivide $[a, b]$ into n subinterval whose widths are $\Delta x_1, \Delta x_2, \Delta x_3, \dots, \Delta x_n$

The subintervals are said to partition the interval, and the largest subinterval width is called the mesh size of the partition. This is denoted by $\max_k \Delta x_k$

Read as "maximum of the $k \Delta x_k$."

In this figure



We see that

If $[a, b]$ is partitioned into n subintervals, and if x_k^* is an arbitrary point in k^{th} subinterval, then $f(x_k^*) \Delta x_k$ is the area of the rectangle with height $f(x_k^*)$ and width Δx_k ,

and $\sum_{k=1}^n f(x_k^*) \Delta x_k$ is the sum of the shaded rectangular areas.

Now if we let $\max_k \Delta x_k \rightarrow 0$ the width of EVERY rectangle tends to 0 because none of them exceeds the max. So we have the area under the curve now defined more generally as

$$A = \lim_{\max_k \Delta x_k \rightarrow 0} \sum_{k=1}^n f(x_k^*) \Delta x_k$$

DEFINITION 5.6.1 (Area under a curve)

If the function f is continuous on $[a, b]$ and if $f(x) \geq 0$ for all x in $[a, b]$, then the area under the curve $y = f(x)$ over the interval $[a, b]$ is defined by

$$A = \lim_{\max_k \Delta x_k \rightarrow 0} \sum_{k=1}^n f(x_k^*) \Delta x_k$$

2) DEFINITE INTEGRAL OF CONTINUOUS FUNCTIONS WITH NONNEGATIVE VALUES

The limit in the definition we just saw is VERY important. It has special notation. We write it as

$$\int_a^b f(x) dx = \lim_{\max_k \Delta x_k \rightarrow 0} \sum_{k=1}^n f(x_k^*) \Delta x_k$$

The expression on the left is called the definite integral of f from a to b . a and b are called the upper and lower limits of integration respectively.

There is a relationship btw this definite integral and the indefinite integral we talked about earlier.

We will see it later.

So with this notation we can say that

$$\int_a^b f(x) dx = \text{Area under the curve } y = f(x) \text{ over } [a, b].$$

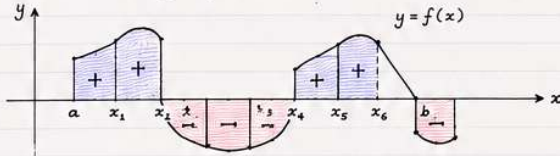
Our goal will be to evaluate the definite integral efficiently instead of using the definition all the time.

Lecture #29 THE DEFINITE INTEGRAL (Part-2)

3) DEFINITE INTEGRAL OF CONTINUOUS FUNCTIONS WITH NEGATIVE AND POSITIVE VALUES

Now we want to extend our area definition to include continuous functions on $[a, b]$ that have both positive and negative values on $[a, b]$.

Look at the following figure



The rectangles below $[a, b]$ but above the curve $f(x)$ have some area just the way their counterparts above $[a, b]$ but below curve $f(x)$ do. The difference is that we can view the number representing the area of the rectangles below as negative values of $f(x)$ for some x_k^* .

So the Riemann sum gives

$$\begin{aligned} \sum_{k=1}^6 f(x_k^*) \Delta x_k &= A_1 + A_2 - A_3 - A_4 + A_5 - A_6 \\ &= (A_1 + A_2 + A_5) - (A_3 + A_4 + A_6) \\ &= (\text{area above } x\text{-axis}) - (\text{area below } x\text{-axis}) \end{aligned}$$

This says that the Riemann sum is the difference of two areas: the total area of rectangles above the x -axis, minus the total area of rectangles below the x -axis.

Now as $\max \Delta x_k$ goes to 0, the situation still works out nicely with the large # of rectangles filling in any left over space between the earlier rectangles and the curve.

So we have the following definition.

DEFINITION 5.6.2 (Net Signed Area)

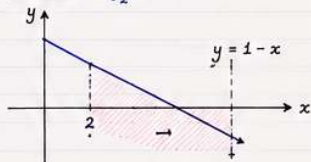
If the function f is continuous on $[a, b]$, and can assume both positive and negative values, then the net signed area A between $y = f(x)$ and the interval $[a, b]$ is defined by

$$\begin{aligned} A &= \lim_{\max \Delta x_k \rightarrow 0} \sum_{k=1}^n f(x_k^*) \Delta x_k \\ &= \int_a^b f(x) dx \end{aligned}$$

Net signed area means that the total difference of the two areas above and below may be negative.

If that happens then we just treat the negative as representing the fact that the area below is larger than the area above.

EXAMPLE 2 Evaluate $\int_2^4 (1-x) dx$



Geometrically, this region lies below the interval $[2, 4]$. The region is a trapezoid and we can find the area using its dimensions.

$$\begin{aligned} \text{Base 1 (at } x=2) &= y(2) = 1-2 = -1 \Rightarrow |b_1| = 1 \\ \text{Base 2 (at } x=4) &= y(4) = 1-4 = -3 \Rightarrow |b_2| = 3 \\ \text{Height } h &= 4-2 = 2 \end{aligned}$$

$$\text{Area (magnitude)} = \frac{1}{2} h (|b_1| + |b_2|) = \frac{1}{2} (2) (1+3) = 4$$

Since region is below x -axis, net signed area is negative.

$$\int_2^4 (1-x) dx = -4$$

4) DEFINITE INTEGRAL OF FUNCTIONS WITH DISCONTINUITIES

For the two definitions we have seen so far, f was a continuous function. This guaranteed that the area defined as a limit could always be found since the limit exists. Now what if f has a finite number of discontinuities on $[a, b]$?

In that case we break $[a, b]$ at the points of discontinuity and apply the definition on each subinterval where f is continuous.

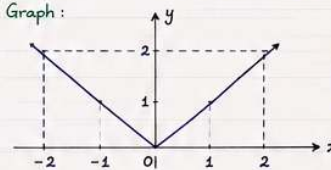
Then we add the results.

Hence the definite integral exists for piecewise continuous functions on $[a, b]$ as well.

EXAMPLE 3 $\int_{-2}^2 |x| dx$

The function $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

Graph:



We break at $x=0$ (point of change in expression).

$$\begin{aligned} \int_{-2}^2 |x| dx &= \int_{-2}^0 (-x) dx + \int_0^2 x dx \\ &= \left[-\frac{x^2}{2} \right]_{-2}^0 + \left[\frac{x^2}{2} \right]_0^2 \\ &= \left(0 - \left(-\frac{4}{2} \right) \right) + \left(\frac{4}{2} - 0 \right) \\ &= 2 + 2 = 4 \end{aligned}$$

5) PROPERTIES OF THE DEFINITE INTEGRAL

- (i) $\int_a^a f(x) dx = 0$
- (ii) $\int_a^b f(x) dx = - \int_b^a f(x) dx$
- (iii) $\int_a^b k dx = k(b-a)$ (k is constant)
- (iv) $\int_a^b [f(x) + g(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx$
- (v) If $a \leq c \leq b$, then $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$

EXAMPLE 4 (Additive Property)

Use property (v) to find $\int_1^3 f(x) dx$, given that $\int_1^2 f(x) dx = 5$ and $\int_2^3 f(x) dx = -2$

Solution:

Here $a = 1$, $c = 2$, $b = 3$.

By property (v),

$$\begin{aligned} \int_1^3 f(x) dx &= \int_1^2 f(x) dx + \int_2^3 f(x) dx \\ &= 5 + (-2) = 3 \end{aligned}$$

6) INEQUALITIES INVOLVING DEFINITE INTEGRALS

- (i) If $f(x) \leq g(x)$ on $[a, b]$, then $\int_a^b f(x) dx \leq \int_a^b g(x) dx$
- (ii) If $f(x) \geq 0$ on $[a, b]$, then $\int_a^b f(x) dx \geq 0$
- (iii) If $m \leq f(x) \leq M$ on $[a, b]$, then $m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$

EXAMPLE 5

Show that $\int_0^1 (x^3 - 25) \cos x dx$ is negative.

Solution:

On $[0, 1]$, we have $\cos x \geq \cos 1 > 0$.

Also $x^3 - 25 < 0$ (since $x^3 \leq 1$).

So $(x^3 - 25) \cos x \leq (\text{negative}) \times (\text{positive}) < 0$ on $[0, 1]$.

Hence, by property (ii), $\int_0^1 (x^3 - 25) \cos x dx < 0$.

(Negative)

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Lecture # 30

First Fundamental Theorem of Calculus

- * 1st fundamental theorem of calculus.
- * Relationship between definite and indefinite integrals.
- * Mean Values theorem for Integrals.
- * Average Values of a function.

This is the 1st fundamental theorem of calculus.

Theorem 5.7.1 (The first fundamental theorem of calculus)

If f is continuous on $[a, b]$ and if F is an antiderivative of f on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a)$$

This theorem tells us how to evaluate EASILY the definite integral.

It says that find the anti-derivative of $f(x)$ first, call it $F(x)$, and then evaluate this function on the limits of the integration. Let's prove this.

Proof

We will use the Mean Value Theorem involving derivatives to prove the first fundamental theorem of calculus.

$$\int_a^b f(x) dx = F(b) - F(a)$$

Let us subdivide the given interval of integration namely $[a, b]$ into n subintervals using the points $x_1, x_2, \dots, x_{n-1}$ in the interval $[a, b]$ such that

$$a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b.$$

So now we have n subintervals of $[a, b]$. We can denote the widths of these intervals as $\Delta x_1, \Delta x_2, \dots, \Delta x_n$. Where for example $\Delta x_2 = x_2 - x_1$.

Since $F'(x) = f(x)$ for all x in $[a, b]$, it is obvious that $F(x)$ satisfies the requirements of the Mean Value Theorem involving derivatives on each of the subintervals. So by MVT, we can find points $x_1^*, x_2^*, \dots, x_n^*$. In each of the respective subintervals $[a, x_1], [x_1, x_2], [x_2, x_3], \dots, [x_{n-1}, b]$.

So we now make the following equations.

- * $F(x_1) - F(a) = F(x_1) - F(x_0) = F'(x_1^*)(x_1 - x_0) = f(x_1^*)\Delta x_1 \quad \text{--- (1)}$
- * $F(x_2) - F(x_1) = F(x_2) - F(x_1) = F'(x_2^*)(x_2 - x_1) = f(x_2^*)\Delta x_2 \quad \text{--- (2)}$
- * $F(x_3) - F(x_2) = F(x_3) - F(x_2) = F'(x_3^*)(x_3 - x_2) = f(x_3^*)\Delta x_3 \quad \text{--- (3)}$
- ...
- ...
- ...
- * $F(b) - F(x_{n-1}) = F(x_n) - F(x_{n-1}) = F'(x_n^*)(x_n - x_{n-1}) = f(x_n^*)\Delta x_n \quad \text{--- (n)}$

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These are easy if you remember that this notation and the definite integral are the same thing!!

Relationship between definite and indefinite integrals

In applying the 1st theorem of Calc, it does not matter WHICH anti-derivative of f is used.

If F is any anti-derivative of f then all the others have form $F(x) + C$ by theorem 5.2.2.

Hence we have the following:

$$\begin{aligned} [F(x) + C]_a^b &= [F(x)]_a^b + [C]_a^b = [F(x)]_a^b + (C - C) \\ &= F(b) - F(a) = \int_a^b f(x) dx \quad \dots (A) \end{aligned}$$

This shows that all anti-derivatives of f on $[a, b]$ give the same values for the definite integral.

Now since $\int_a^b f(x) dx = F(b) - F(a)$

It follows from (A) that

$$\int_a^b f(x) dx = \int_a^b f(x) dx \quad (\text{same value})$$

So we can first evaluate the indefinite integral and use the limits as well to evaluate the definite integral.

This relates the definite and the indefinite integrals. © Copyright Virtual University of Pakistan | 203

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Adding up these equations we get

$$F(b) - F(a) = \sum_{k=1}^n f(x_k^*) \Delta x_k \quad \text{--- (*)}$$

Now increase n in such a way that

$$\max \Delta x_k \rightarrow 0$$

Since f is assumed to be continuous, we have the following result

$$F(b) - F(a) = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x_k = \int_a^b f(x) dx = \left\{ \begin{array}{l} \text{THEOREM} \\ 5.6.9(a) \end{array} \right.$$

I did not apply the limits on the left because even if I did, the expression does not involve n and therefore nothing happens.

$F(b) - F(a)$ can also be written as $[F(x)]_a^b$, so we have

$$\int_a^b f(x) dx = [F(x)]_a^b$$

Example

Evaluate $\int_1^2 x dx$

The function $F(x) = \frac{x^2}{2}$ is an antiderivative of $f(x) = x$. So we have

$$\int_1^2 x dx = \left[\frac{x^2}{2} \right]_1^2 = \frac{2^2}{2} - \frac{1^2}{2} = \frac{4}{2} - \frac{1}{2} = \frac{3}{2}$$

Here are a few properties of the bracket notation we just saw.

Prove these yourself!

Properties

1. $[c F(x)]_a^b = c [F(x)]_a^b$
2. $[F(x) + G(x)]_a^b = [F(x)]_a^b + [G(x)]_a^b$
3. $[F(x) - G(x)]_a^b = [F(x)]_a^b - [G(x)]_a^b$

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Example

Using the 1st theorem of calculus, find the area under the curve $y = \cos(x)$ over the interval $[0, 2\pi]$.

Since $\cos(x) \geq 0$ for $0 \leq x \leq \pi$, the area is

$$A = \int_0^{2\pi} \cos(x) dx = [\sin(x)]_0^{2\pi} = \sin(2\pi) - \sin(0) = 0 - 0 = 0$$

I deliberately chose $C = 0$ since we just saw that the value of $C = 0$ doesn't change the answer.

Example

Evaluate $\int_0^3 (x^2 - 4x + 1) dx$

$$\begin{aligned} \int_0^3 (x^2 - 4x + 1) dx &= \int_0^3 x^2 dx - \int_0^3 4x dx + \int_0^3 1 dx \\ &= \left[\frac{x^3}{3} \right]_0^3 - 4 \left[\frac{x^2}{2} \right]_0^3 + [x]_0^3 \\ &= \left[\frac{27}{3} - 0 \right] - 4 \left[\frac{9}{2} - 0 \right] + [3 - 0] \\ &= 9 - 18 + 3 = -6 \end{aligned}$$

Example

Evaluate $\int_0^6 f(x) dx$ where $f(x) = \begin{cases} x^2, & \text{if } x < 2 \\ 3x - 2, & \text{if } x \geq 2 \end{cases}$

From theorem 5.6.6

$$\begin{aligned} \int_0^6 f(x) dx &= \int_0^2 f(x) dx + \int_2^6 f(x) dx = \int_0^2 x^2 dx + \int_2^6 (3x - 2) dx \\ &= \left[\frac{x^3}{3} \right]_0^2 + \left[\frac{3x^2}{2} - 2x \right]_2^6 = \left(\frac{8}{3} - 0 \right) + \left(\frac{108}{2} - 12 - \frac{12}{2} + 4 \right) \\ &= \frac{8}{3} + (54 - 12 - 6 + 4) = \frac{8}{3} + 40 = \frac{128}{3} \end{aligned}$$

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Lecture # 31 Evaluating Definite Integral by Substitution.

Evaluating Definite Integrals.

- By substitution
- Approximation by Riemann Sums

Evaluating definite integral by substitution

We want to evaluate the following definite integral by substitution

$$\int_a^b f(x) dx$$

Method 1

Evaluate the indefinite integral $\int f(x) dx$ by substitution, and then use the relationship.

$$\int_a^b f(x) dx = \left[\int f(x) dx \right]_a^b$$

Method 2

First represent the definite integral in the form

$$\int_a^b f(x) dx = \int_a^b h(g(x)) g'(x) dx$$

And then make the substitution $u = g(x)$, $du = g'(x) dx$ directly into the definite integral.

But now we have to change integration limits from x -limits to u -limits as follows

$$\begin{cases} u = g(a), & \text{if } x = a \\ u = g(b), & \text{if } x = b \end{cases}$$

This leaves us with a new integral in terms of u

$$\int_a^b f(x) dx = \int_{g(a)}^{g(b)} h(u) du$$

This will be simpler if the choice of u is good.

Example 1 Evaluate $\int_0^2 (3x^2 + 2x) dx$

Solution:

Method 1

Let $u = x^3 + x^2$, $du = (3x^2 + 2x) dx$

$$\begin{aligned} \text{So } \int (3x^2 + 2x) dx &= \int du = u + C \\ &= x^3 + x^2 + C \end{aligned}$$

So,

$$\begin{aligned} \int_0^2 (3x^2 + 2x) dx &= \left[x^3 + x^2 \right]_0^2 \\ &= (2^3 + 2^2) - (0^3 + 0^2) \\ &= (8 + 4) - 0 = 12 \end{aligned}$$

Method 2

Let $u = x^3 + x^2$.

$$\begin{aligned} \text{If } x = 0, \\ u &= 0^3 + 0^2 = 0 \end{aligned}$$

$$\begin{aligned} \text{If } x = 2, \\ u &= 2^3 + 2^2 = 8 + 4 = 12 \end{aligned}$$

Thus,

$$\begin{aligned} \int_0^2 (3x^2 + 2x) dx &= \int_0^{12} 1 du \\ &= \left[u \right]_0^{12} = 12 - 0 = 12 \end{aligned}$$

Approximation by Riemann Sums

Recall that a Riemann Sum is the expression

$$\sum_{k=1}^n f(x_k^*) \Delta x_k$$

which occurs when we try to approximate the area under the curve.

- If we take the limits of this sum, we get the definite integral.
- If we don't take the limits, then we can get a good approximation to the definite integral if n is relatively LARGE.

So we can say that for large n

$$\int_a^b f(x) dx \approx \sum_{k=1}^n f(x_k^*) \Delta x_k$$

We need to use this approx when it's impossible to evaluate exactly.

It is a good idea in this case to use a regular partition of the interval $[a, b]$ which gives same width for each subinterval.

So we then write

$$\int_a^b f(x) dx \approx \Delta x [f(x_1^*) + f(x_2^*) + \dots + f(x_n^*)]$$

where $\Delta x = \frac{b-a}{n}$.

This formula produces a left endpoint approximation, a right endpoint approximation and midpoint approximation, depending on the choice of the point x_k^* .

Example 2 Evaluate $\int_0^\pi \cos(x - \frac{\pi}{4}) dx$

Let $u = x - \frac{\pi}{4}$ so that $du = dx$.

Also $u = -\frac{\pi}{4}$ if $x = 0$, $u = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$ if $x = \pi$.

So,

$$\begin{aligned} \int_0^\pi \cos(x - \frac{\pi}{4}) dx &= \int_{-\pi/4}^{3\pi/4} \cos u du = \left[\sin u \right]_{-\pi/4}^{3\pi/4} \\ &= \sin\left(\frac{3\pi}{4}\right) - \sin\left(-\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} - \left(-\frac{\sqrt{2}}{2}\right) = \sqrt{2} \end{aligned}$$

Example 3 Approximate $\int_0^\pi \sqrt{1-x^2} dx$ using the left endpoint, right end point, and the midpoint approx each with $n=10$, $n=20$, $n=50$ and $n=100$ subintervals.

Solution:

n	Left Endpoint	Right Endpoint	Midpoint
10	0.803957	0.767169	0.785392
20	0.793279	0.777635	0.785398
50	0.789545	0.781227	0.785398
100	0.787455	0.783343	0.785398

The exact value of the integral is $\frac{\pi}{4}$. This is consistent with the preceding computations, since $\frac{\pi}{4} \approx 0.785398163$.

Lecture # 32SECOND FUNDAMENTAL THEOREM OF CALCULUSOUTLINE :

- ★ Dummy Variable
- ★ Definite Integrals with variable upper limit
- ★ Second fundamental theorem of Calculus
- ★ Existence of Ant derivatives for continuous functions
- ★ Functions defined by Integrals

1) DUMMY VARIABLE

Here is some notational stuff.

If we change the letter for the variable of integration but don't change the limits, then the values of the definite integral are unchanged.

That is to say

$$\int_a^b f(x) dx = \int_a^b f(t) dt = \int_a^b f(y) dy$$

All have the same value.

For this reason, we call the letter used for the variable of integration DUMMY variable.

Example :

Evaluate $\int_1^3 (2x+6) dx$, $\int_1^3 (2t+6) dt$, $\int_1^3 (2y+6) dy$

Solution :

$\begin{aligned} & \int_1^3 (2x+6) dx \\ &= [x^2 + 6x]_1^3 \\ &= [(3)^2 + 6(3)] - [(1)^2 + 6(1)] \\ &= (9+18) - (1+6) \\ &= 27 - 7 = \boxed{20} \end{aligned}$	$\begin{aligned} & \int_1^3 (2t+6) dt \\ &= [t^2 + 6t]_1^3 \\ &= [9 + 18] - [1 + 6] \\ &= 27 - 7 = \boxed{20} \end{aligned}$	$\begin{aligned} & \int_1^3 (2y+6) dy \\ &= [y^2 + 6y]_1^3 \\ &= [9 + 18] - [1 + 6] \\ &= 27 - 7 = \boxed{20} \end{aligned}$
--	--	--

Hence all three integrals have the same value = 20.

2) DEFINITE INTEGRALS WITH VARIABLE UPPER LIMIT OF INTEGRATION

We will now consider definite integrals of the form $\int_2^x f(t) dt$ where the upper limit is a variable rather than a number.

In such integrals, we will use a different letter for the integration variable.

This will distinguish between the limit of integration and the variable of integration.

Let's do an example :

Example :

Evaluate $\int_2^x t^2 dt$

Solution :

$$\begin{aligned} \int_2^x t^2 dt &= \left[\frac{t^3}{3} \right]_2^x \\ &= \frac{x^3}{3} - \frac{2^3}{3} = \boxed{\frac{x^3 - 8}{3}} \end{aligned}$$

Note that the result in this case is a FUNCTION of x .

In all such integrals with an x as a limit, the final result is a function of x .

3) SECOND FUNDAMENTAL THEOREM OF CALCULUS

In lecture 25, we say that if f is a nonnegative continuous function over $[a, b]$, and $A(x)$ represents the area under the curve $y = f(x)$ over the interval $[a, x]$ as a function of x , then

$$A'(x) = f(x) \quad (\text{Derivative of Area function is the function } f \text{ itself})$$

Lecture # 32 SECOND FUNDAMENTAL THEOREM OF CALCULUS (Part-2)

4) SECOND FUNDAMENTAL THEOREM OF CALCULUS (CONTINUED)

Now let's write $A(x)$ as a definite integral since that's how it is represented.

$$A(x) = \int_a^x f(t) dt \quad (\text{Area function})$$

If we now take the derivative w.r.t. x , we get

$$\frac{d}{dx} [A(x)] = \frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x)$$

This result holds for all continuous functions, and we can state it as a theorem.

THEOREM 5.9.1 (Second Fundamental Theorem of Calculus)

If f is continuous on an interval I and a is any number in I , then

$$\frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x) \quad \text{for all } x \in I$$

Read as :

- ① If the integrand is continuous, then the derivative of a definite integral with respect to its upper limit is equal to the integrand evaluated at the upper limit.
- ② The derivative of a function representing the area under the curve of another continuous function f is equal to the function f on a given interval.

Example 2 :

Since $f(x) = 3x^2$ is continuous everywhere, evaluate $\frac{d}{dx} \left[\int_1^x 3t^2 dt \right]$ to check the validity of 2nd fundamental theorem of calculus.

Solution :

By 2nd fundamental theorem of calculus,

$$\frac{d}{dx} \left[\int_1^x 3t^2 dt \right] = 3x^2$$

We check it by evaluating the integral.

$$\int_1^x 3t^2 dt = [t^3]_1^x = x^3 - 1^3 = x^3 - 1$$

$$\text{Differentiating this gives } \frac{d}{dx} (x^3 - 1) = 3x^2 \\ = 3x^2 \quad (\text{Verified})$$

5) EXISTENCE OF ANTI-DERIVATIVES OF CONTINUOUS FUNCTIONS

If f is a continuous function on an interval I and ' a ' is any point in I , then this formula

$$\frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x) \quad \text{tells us that } F(x) = \int_a^x f(t) dt \quad \text{is an}$$

anti-derivative of f on I . So, now with the 2nd theorem of calculus, we can say that

EVERY continuous function on an interval has an anti-derivative on that interval.

6) FUNCTIONS DEFINED BY INTEGRALS

So far we have been able to determine the anti-derivative of a given function by integration techniques.

But it's not always possible to find the anti-derivative of a given function f which is continuous on an interval and thus is guaranteed to have an anti-derivative on that interval.

If this is the case, we get functions (anti-derivatives) that are defined in terms of integrals and nothing simpler that we have seen so far.

Example 3 :

$$\text{Evaluate } \int_1^3 \frac{1}{x} dx$$

Solution :

This function is continuous on $[1, 3]$ so it is integrable on $[1, 3]$. By the 2nd theorem of calculus, an antiderivative is

$$F(x) = \int_1^x \frac{1}{t} dt$$

Using the 1st theorem of calculus we can attempt to evaluate this integral.

$$F(3) - F(1) = \int_1^3 \frac{1}{t} dt = \left[\int_1^3 \frac{1}{t} dt \right] = \int_1^3 \frac{1}{t} dt \\ = \int_1^3 \frac{1}{t} dt - \int_1^1 \frac{1}{t} dt = \int_1^3 \frac{1}{t} dt - \int_1^1 \frac{1}{t} dt \\ = \int_1^3 \frac{1}{t} dt \quad (\text{Cannot be simplified further})$$

Hence, this formula $\int_1^x \frac{1}{t} dt$ defines a FUNCTION in terms of integrals.

- But this is not a function like any we have seen so far.
- In fact, $1/t$ cannot be integrated using polynomials, rational functions, or any other function we have seen.
- But we can approximate this function by using numerical methods like Riemann sums.

Exam Tip :

Remember : Upper limit is the variable, lower limit is constant.

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Lecture # 33

Application to the Definite Integral

Application of the Definite Integral

- * Area problem: Area Between two Curves
- * Area between $y = f(x)$ and $y = g(x)$
- * Area between $x = v(y)$ and $x = w(y)$

First Area problem

Area between two curves:

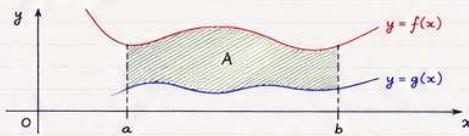
Area between $y = f(x)$ and $y = g(x)$

Suppose that f and g are continuous functions on an interval $[a, b]$ and

$$f(x) \geq g(x) \text{ for } a \leq x \leq b$$

(This means that the curve $y = f(x)$ is above $y = g(x)$ and that the two can touch but not cross).

Find the area A of the region bounded above by $y = f(x)$, below by $y = g(x)$, and on the sides by the lines $x = a$ and $x = b$.



If f and g are nonnegative on $[a, b]$ then we have

$$A = [\text{area under } f] - [\text{area under } g]$$

In terms of integrals we can say this as

$$A = \int_a^b f(x) dx - \int_a^b g(x) dx = \int_a^b [f(x) - g(x)] dx$$

What if f and g take on negative values too?

This means that their graphs are below the x -axis for some values in $[a, b]$.

This can be remedied if we translate the two graphs by a constant m so big that it shifts both f and g above the x -axis.

This shift does not affect the area between the two curves.

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$$A = \int_a^b [f(x) + m] dx - \int_a^b [g(x) + m] dx = \int_a^b [f(x) - g(x)] dx$$

The constants cancel each other in the calculations.

Here is the formal definition.

DEFINITION 6.1 (Area Formula)

If f and g are continuous functions on the interval $[a, b]$, and $f(x) \geq g(x)$ for all x in $[a, b]$, then the area of the region bounded above by $y = f(x)$, below by $y = g(x)$, on the left by the line $x = a$, and on the right by the line $x = b$ is

$$A = \int_a^b [f(x) - g(x)] dx$$

A few things to keep in mind.

- * If the region confined by two curves is complicated, then it may require some careful thought to determine the integrand.
- * It may be hard to find the limits of integration.
- * Here is a procedure we can follow.

1. Sketch the region and the two bounding curves.
2. Find the points of intersection (if any).
3. Determine which curve is on top (f) and which is on the bottom (g) on the interval of interest.
4. Find the limits of integration (left and right endpoints).
5. Compute $A = \int_a^b [f(x) - g(x)] dx$.

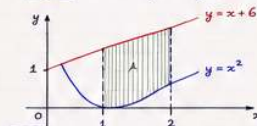
Example

Find the area of the region bounded above by $y = x + 6$, and below by $y = x^2$ and on the sides by lines $x = 0$ and $x = 2$.

Solution:

The region and a vertical line through it are shown in this figure.

The line extends from $f(x) = x + 6$ on the top to $y = x^2$ at the bottom.



As the line moves left to right, its leftmost position will be $x = 0$ and right most will be $x = 2$

$$A = \int_0^2 [(x+6) - x^2] dx = \int_0^2 (6+x-x^2) dx$$

$$= \left[6x + \frac{x^2}{2} - \frac{x^3}{3} \right]_0^2 = \left[12 + 2 - \frac{8}{3} \right] - 0 = \frac{34}{3}$$

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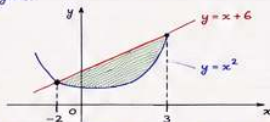
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Example

Find the area of the region that is enclosed between $y = x^2$ and $y = x + 6$.

Solution:

There are no lines at the right or the left. So how do we find the limits of integration? Well here is a sketch of the region.



Note that the integration limits are defined by the points where the two curves intersect.

We can find these points by equating the two equations and solving for x .

We get

$$x^2 = x + 6 \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x-3)(x+2) = 0$$

$$\Rightarrow x = 3 \text{ and } x = -2$$

So now we have

$$A = \int_{-2}^3 [(x+6) - x^2] dx = \int_{-2}^3 (6+x-x^2) dx$$

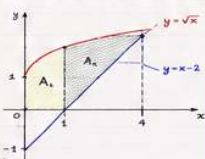
$$= \left[6x + \frac{x^2}{2} - \frac{x^3}{3} \right]_{-2}^3 = \left[18 + \frac{9}{2} - 9 \right] - \left[-12 + 2 - \frac{8}{3} \right] = \frac{21}{2} + \frac{22}{3} - \frac{63}{6} + \frac{44}{6} = \frac{125}{6}$$

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This causes us problems when we try to determine $f(x) - g(x)$ for the integration.

Instead we will divide the region into two parts and evaluate the area of each region and then add together.

Here is how we will divide it.



$$A_1: f(x) = \sqrt{x}, g(x) = -\sqrt{x}, a = 0, b = 1$$

$$A_1 = \int_0^1 [\sqrt{x} - (-\sqrt{x})] dx = \int_0^1 2\sqrt{x} dx$$

$$= 2 \left[\frac{2}{3} x^{3/2} \right]_0^1 = \left[\frac{4}{3} x^{3/2} \right]_0^1 = \frac{4}{3}$$

$$A_2: f(x) = \sqrt{x}, g(x) = x-2, a = 1, b = 4$$

$$A_2 = \int_1^4 [\sqrt{x} - (x-2)] dx = \int_1^4 (\sqrt{x} - x + 2) dx$$

$$= \left[\frac{2}{3} x^{3/2} - \frac{x^2}{2} + 2x \right]_1^4 = \left(\frac{16}{3} - 8 + 8 \right) - \left(\frac{2}{3} - \frac{1}{2} + 2 \right)$$

$$= \left(\frac{16}{3} \right) - \left(\frac{13}{6} \right) = \frac{32}{6} - \frac{13}{6} = \frac{19}{6}$$

$$\text{Total area of region} = A_1 + A_2 = \frac{4}{3} + \frac{19}{6} = \frac{8}{6} + \frac{19}{6} = \frac{27}{6} = \frac{9}{2}$$

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Example

Find the area between $x = v(y)$ and $x = w(y)$ by $y = \sqrt{x}$ and $y = x - 2$.

Solution:

First we need to know where the two curves intersect.

Let's write down $y = x - 2$ as $x = y + 2$ and then equate $y = \sqrt{x}$ and $x = y + 2$.

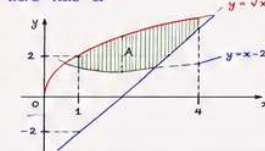
This gives

$$y^2 = y + 2 \Rightarrow y^2 - y - 2 = 0 \Rightarrow (y-2)(y+1) = 0$$

$$\Rightarrow y = -1 \text{ and } y = 2$$

Substituting these into either equation will give us the desired x -values, which are $x = 1$ and $x = 4$ respectively.

Here are x^2



From this figure, we see that the upper boundary ($f(x)$) is defined by $y = \sqrt{x}$.

The lower boundary has two pieces:

- 1) $y = -\sqrt{x}$ for x between -1 and 0
- 2) $y = x - 2$ for x between 1 and 4

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Example

Find the area of the region confined by $y = \sqrt{x}$ and $y = x - 2$ or $x = y + 2$.

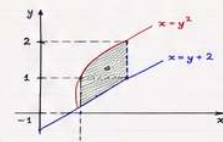
Solution:

Left boundary is $x = y^2$
Right boundary is $x = y + 2$
Limits are from -1 to 2

$$A = \int_{-1}^2 [(y+2) - y^2] dy = \int_{-1}^2 (-y^2 + y + 2) dy$$

$$= \left[-\frac{y^3}{3} + \frac{y^2}{2} + 2y \right]_{-1}^2 = \left(-\frac{8}{3} + 2 + 4 \right) - \left(\frac{1}{3} + \frac{1}{2} - 2 \right)$$

$$= \left(\frac{10}{3} \right) - \left(-\frac{7}{6} \right) = \frac{20}{6} + \frac{7}{6} = \frac{27}{6} = \frac{9}{2}$$



The area is shown in the figure above.

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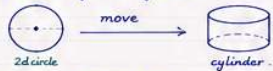
Volume by Slicing; Disks and Washers

- * Definite Integrals to find Volumes of three dimensional solids
- * Cylinders
- * The method of Slicing
- * Volumes by cross sections perpendicular to the x axis
- * Volumes by cross sections perpendicular to the y axis
- * Volumes of solids of revolution by:
 - i. Volumes by Disks perpendicular to x axis
 - ii. Volumes by Disks perpendicular to y axis
 - iii. Volumes by washers perpendicular to x axis
 - iv. Volumes by washers perpendicular to y axis

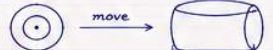
Cylinders

How to go from 0-d to 3-d and beyond in terms of a cube. We can get other 3d solids by moving 2d planes in a certain fashion in 3d space.

- If I move a 2d circle along a line that is perpendicular to the circle, I will get a cylinder.



- If I move a circle with a hole in it, or a washer, along a line perpendicular to the washer, then I will get a cylinder with a hole in it.



- In general, if I move a 2d plane in a direction along a line perpendicular to the region, then I get a **RIGHT CYLINDER**.



Every solid object has Volume, Volume is the 3d area!

We can find the volume of a right cylinder generated by moving a 2d plane by the following formula

$$V = A \cdot h$$

Where A is the area of the 2d object, and h is the height of the cylinder.

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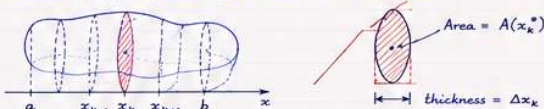
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Let's call the area of a cross section at some arbitrary point x $A(x)$

Now Let's divide the interval $[a, b]$ into n subintervals of width $\Delta x_1, \Delta x_2, \dots, \Delta x_n$ by inserting the points $x_1, x_2, \dots, x_n$ between a and b .

We can now pass a plane perpendicular to the x -axis through each of these points.

These planes will subdivide the solid into SLICES $S_1, S_2, \dots, S_n$. Consider a typical slice S_k as shown in the figure.



If this slice is very thin, then the cross section of this slice S_k will not vary too much in the interval Δx_k which defines the thickness of the slice S_k . That is, the cross section will have the same shape in the interval if Δx_k is thin.

So let's choose a point x_k^* in the interval Δx_k and take a cross section of the slice at this point. Then the area of this cross section will be $A(x_k^*)$.

The volume of the slice S_k then will be approximately

$$V_k \approx A(x_k^*) \Delta x_k$$

We can use this formula for all the THIN slices we make to get their respective volumes.

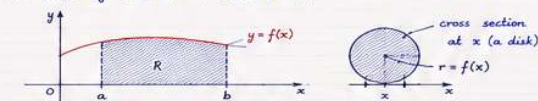
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I. Disks perpendicular to x -axis

Let f be a nonnegative and continuous function on $[a, b]$.

Let R be the region bounded above by the graph of f and on the sides by lines $x = a$ and $x = b$.



If this solid is revolved around the x -axis, it generates a solid with a circular cross sections.

The cross section at x has radius $f(x)$ so the area of this cross section is $A(x) = \pi[f(x)]^2$.

So from the formulas we just saw we can write for the volume of this solid

So the Volume of a cylinder is the area of a cross section of the cylinder multiplied by the height of the cylinder.

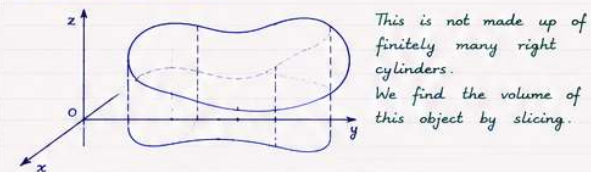
But there are 3d solids that are not right cylinders and are neither made up of finitely many right cylinders.

So we cannot use the formulas for the area of the right cylinders to find the area of such an object.

In this case we use the technique of slicing.

The method of Slicing

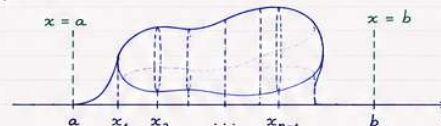
Here is such a 3d solid



This is not made up of finitely many right cylinders. We find the volume of this object by slicing.

Impose an x axis on the solid.

Then we can imagine that the solid is bounded to the left by a plane perpendicular to the x -axis at $x = a$ and to the right by a plane at $x = b$.



The problem with finding the volume of this non-cylindrical solid is clear now: at any point along the x -axis, the cross sections perpendicular to the x -axis will have different areas!

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The total volume of the solid can be approximated by

$$V \approx V_1 + V_2 + \dots + V_n \approx \sum_{k=1}^n A(x_k^*) \Delta x_k$$

Now increase the number of slices. Then the slices will become VERY THIN and the approx will get better and better. So we can take the limit to get

$$V = \lim_{\max \Delta x_k \rightarrow 0} \sum_{k=1}^n A(x_k^*) \Delta x_k = \int_a^b A(x) dx$$

- * Volumes by cross sections perpendicular to the x axis
- * Volumes by cross sections perpendicular to the y axis
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 - iv. Washers perpendicular to y -axis

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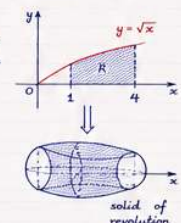
Example 2

Find the volume of solid that is obtained when the region under the curve $y = \sqrt{x}$ over the $[1, 4]$ is revolved about the x -axis.

Solution:

From $V = \pi \int_a^b [f(x)]^2 dx$, the volume is

$$V = \pi \int_1^4 (\sqrt{x})^2 dx = \pi \int_1^4 x dx = \pi \left[\frac{x^2}{2} \right]_1^4 = \pi \left[\frac{16}{2} - \frac{1}{2} \right] = \pi \left(\frac{15}{2} \right) = \frac{15\pi}{2}$$



Example 3

Derive the formula for the volume of a sphere of radius r .

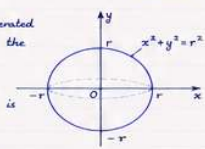
Solution:

As indicated in the figure, a sphere of radius r can be generated by revolving the upper half of the circle $x^2 + y^2 = r^2$ about the x -axis. Since upper half of this circle is graph of

$$y = f(x) = \sqrt{r^2 - x^2},$$

it follows from above formula that the volume of the sphere is

$$V = \pi \int_{-r}^r (r^2 - x^2) dx = \pi \left[r^2 x - \frac{x^3}{3} \right]_{-r}^r = \pi \left[\left(r^3 - \frac{r^3}{3} \right) - \left(-r^3 + \frac{r^3}{3} \right) \right] = \pi \left[\frac{2r^3}{3} + \frac{2r^3}{3} \right] = \frac{4}{3} \pi r^3$$



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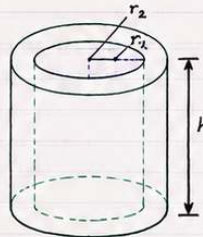
Lecture # 35 VOLUME BY CYLINDRICAL SHELLS

- Cylindrical Shells
- Volume of a Cylindrical Shell
- Cylindrical Shells centered on the y-axis
- Volume through the surface area of a surface created from a cylindrical shell

1) CYLINDRICAL SHELLS

If we take a washer - a disk with a hole in it - and extend it UP, we generate a solid called a cylindrical shell.

This is a solid confined by two concentric right circular cylinders.



2) VOLUME OF A CYLINDRICAL SHELL

The Volume of a cylindrical shell can be expressed as

$$\begin{aligned} V &= (\text{area of cross section}) \cdot (\text{height}) \\ &= \pi r_2^2 h - \pi r_1^2 h \\ &= \pi (r_2^2 - r_1^2) h \\ &= \pi (r_2 + r_1)(r_2 - r_1) h \\ &= 2\pi \left[\frac{r_1 + r_2}{2} \right] (r_2 - r_1) h \\ &= 2\pi (\text{average radius}) \cdot (\text{thickness}) \cdot (\text{height}) \end{aligned}$$

r_2 = outer radius
 r_1 = inner radius
 h = height
 $(r_2 - r_1)$ = thickness

Let's rewrite this rearrangement of the Volume of a cylindrical shell as

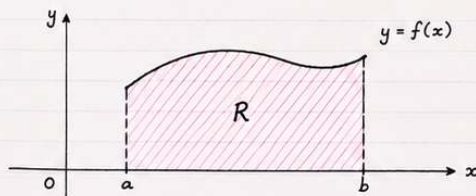
$$V = 2\pi (\text{average radius}) \cdot (\text{height}) \cdot (\text{thickness})$$

Now we can use this formula to compute the Volume of a Solid generated by revolution of a surface around an axis.

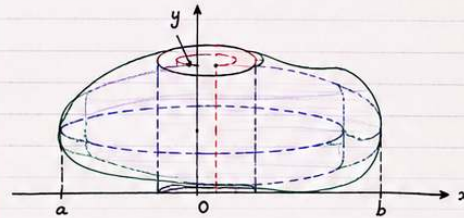
3) SETUP FOR REVOLUTION AROUND THE Y-AXIS

Consider the following:

R is a region bounded by the graph of $f(x)$ on top, below by the x -axis, and to the left by $x = a$, and right by $x = b$.



When we revolve R around the y -axis, we get the solid S as in the picture.



To find the volume of S , here is what we do.

- ★ Subdivide $[a, b]$ into n subintervals with widths $\Delta x_1, \Delta x_2, \dots, \Delta x_n$ by inserting the points $x_1, x_2, \dots, x_n$ between a and b .
- ★ If we draw vertical lines from these points to the graph of f , we get subdivisions $R_1, R_2, \dots, R_n$ of the region R .
- ★ Revolving these strips around the y -axis gives solids $S_1, S_2, \dots, S_n$.

So we can do the following to get the volume of S

$$V(S) = V(S_1) + V(S_2) + \dots + V(S_n)$$

4) APPROXIMATION BY CYLINDRICAL SHELLS

Consider the strip R_k and the corresponding solid S_k .

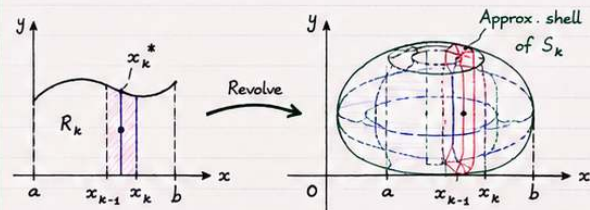
→ The solid S_k is not necessarily a cylindrical shell as it may have a curved upper surface depending on the graph of f .

→ However, if the width Δx_k is very small, then we can get a good approx to the region R_k by the rectangle of width Δx_k and height $f(x_k^*)$ where

$$x_k^* = \frac{x_{k-1} + x_k}{2}$$

is the midpoint of the interval $[x_{k-1}, x_k]$.

→ This in turn will give us a cylindrical shell when we revolve it around y -axis which will be a good approx to the solid.



Lecture # 36 LENGTH OF A PLANE CURVE

1) ARC LENGTH:

We all know how to find the length of a line. In this section, we will develop the ways of finding the length of curves, or lines that twist and turn. The curves we look at will be graphs of functions. The functions we look at will be such that will be continuous on a given interval. Such functions are called smooth functions, and their graphs smooth curves.

So in this lecture, we are concerned with the

ARC LENGTH PROBLEM

Arc Length Problem:

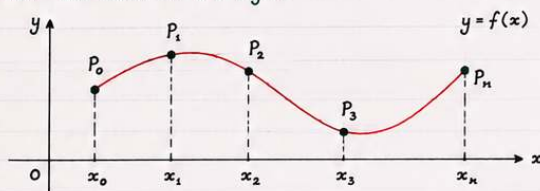
Suppose that f is a smooth function on the interval $[a, b]$. Find the arc length L of the curve $y = f(x)$ over the interval $[a, b]$.

Here question arises that How can we measure the arc length?

Arc length is just the length of a line that is not straight, but can be rounded in various places.

Sometimes we want to measure something with a measuring tape, something that is straight. But often the tape hangs loosely away from the straight object. In such cases our result is not accurate, since if the tape is loose then it will show some extra length. Similarly, if we have a curve surface, we can not measure its length with a straight meter rod accurately.

Consider this curve in the figure



Now say I want to measure the length from P_0 to P_1 .

Let's say I use a straight stick. The stick will be rigid and straight, and so I will miss some length that is curved between the points P_0 and P_1 . If P_0 and P_1 are so close that distance between them become straight then straight stick can be used.

Now I can do the same for all the lengths between all the points in the figure.

The line segments joining the points (P_1 to P_n) form a polygonal path and it's clear that this path is a good approximation of the length of the curve.

Well, it should also be clear that just as we have done so far, if we let the number of these segments joining all points between P_0 to P_n increase, we will get a better approximation to the curve.

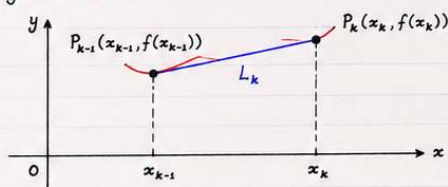
We can do it in the following way

Note in the figure that to each point P_i , there corresponds a point x_i on the x -axis in the interval $[a, b]$

Let's assign the distance between the x_i 's as $\Delta x_1, \Delta x_2, \dots, \Delta x_n$.

Let's look at the k^{th} line segment. We will call it L_k . Here is what it looks like.

Figure 6.4.3



We want to find its length. Using the distance formula, we get

$$L_k = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$$

From the figure 6.4.3, it's clear that

$$\Delta y_k = f(x_k) - f(x_{k-1})$$

So we get

$$L_k = \sqrt{(\Delta x_k)^2 + [f(x_k) - f(x_{k-1})]^2}$$

The Mean Value Theorem (Theorem 4.9.2) can be applied here.

So, we can write our first equation as

$$\begin{aligned} L_k &= \sqrt{(\Delta x_k)^2 + [f(x_k) - f(x_{k-1})]^2} \\ &= \sqrt{(\Delta x_k)^2 + [f'(x_k^*) \Delta x_k]^2} \quad (\text{by M.V.T.}) \\ &= \sqrt{1 + [f'(x_k^*)]^2} \Delta x_k \end{aligned}$$

x_k^* is some point between x_{k-1} and x_k

So, this is the length of one segment.

The length of the WHOLE polygonal path will be

$$L_n = \sum_{k=1}^n L_k = \sum_{k=1}^n \sqrt{1 + [f'(x_k^*)]^2} \Delta x_k$$

Lecture # 37 AREA OF A SURFACE OF REVOLUTION

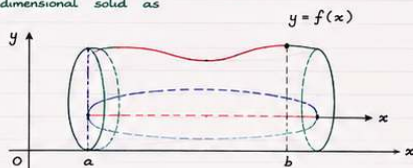
We will study in this lecture

- Definition of Surface Area
- Surface Area Formulas

1) SURFACE AREA PROBLEM:

Let f be a smooth, non negative function on $[a, b]$. Find the area of the surface generated by revolving the portion of the curve $y = f(x)$ between $x = a$ and $x = b$ about the x -axis.

By revolving this curve we get a 3-dimensional solid as



Surface area is roughly the area covered by the surface of a solid in 3-dimensional space.

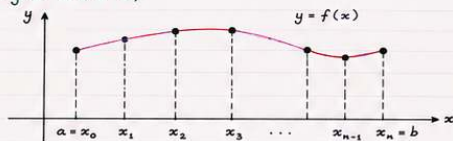
But we need a more precise definition.

Say we have the graph of the function f as stated earlier in 6.5.1 area problem.

As usual, divide the interval $[a, b]$ into subintervals with widths $\Delta x_1, \Delta x_2, \dots, \Delta x_n$ having x -coordinates $a = x_0 < x_1 < x_2 < \dots < x_n = b$.

This in turn subdivides the curve into "sub-curves" which can be approximated by a polygonal path made up of line segments joining the end points of the "sub-curves".

Figure 6.5.2 a)



3) ARC LENGTH OF THE WHOLE POLYGONAL PATH:

The area of the approximated solid is then

$$S_n = \sum_{k=1}^n S_k = \sum_{k=1}^n 2\pi f(x_k^*) \sqrt{(\Delta x_k)^2 + [f'(x_k^*)]^2}$$

As we have done before, let the largest width on the x -axis approach 0 to get

$$S = \lim_{\max \Delta x_k \rightarrow 0} \sum_{k=1}^n 2\pi f(x_k^*) \sqrt{(\Delta x_k)^2 + [f'(x_k^*)]^2} = \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx$$

If we now rotate this polygonal path around the x -axis, we get a solid that is approximately the same in properties to the one we had by rotating the graph of f around the x -axis.

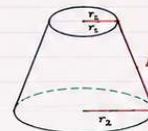
This approximated solid is made up of frustums of a cone.

A frustum of a cone is the solid that you get if you chop off the pointed top of a cone.

It has two radii and a slanted length.

Its surface area is measured by

$$S = \pi (r_1 + r_2) L$$



In our case of the solid, the radii correspond to the height of the points of subdivision on the curve in an interval, and L is the length of the line segment (arc length) in a given interval.

So we can use this formula to find the surface area of each frustum on the approx. solid, and then add them all up.

To get the surface area of the original solid, if we let the number of our subdivisions increase without bound, then we get the surface area of the solid. Here is how it is done.

2) SURFACE AREA OF EACH FRUSTUM:

Let's consider the k -th frustum of our approx. solid.

Its surface area can be found using the formula for the frustum as

$$S_k = \pi [r_1(k) + r_2(k)] L_k = \pi [f(x_{k-1}) + f(x_k)] L_k$$

$r_1 = f(x_{k-1})$ $r_2 = f(x_k)$ L_k is slant length

The formula for L_k is just that of arc length so we can write it as

$$L_k = \sqrt{(\Delta x_k)^2 + [f(x_k^*) - f(x_{k-1})]^2}$$

So

$$S_k = \pi [f(x_{k-1}) + f(x_k)] \sqrt{(\Delta x_k)^2 + [f(x_k^*) - f(x_{k-1})]^2}$$

By the Intermediate Value Theorem, there exists a point x_k^* in the interval $[x_{k-1}, x_k]$ such that

$$f(x_k^*) = \frac{f(x_{k-1}) + f(x_k)}{2}$$

So, we can re-write our equation for S_k as

$$S_k = 2\pi f(x_k^*) \sqrt{(\Delta x_k)^2 + [f'(x_k^*)]^2}$$

4) SURFACE AREA FORMULA:

Let f be a smooth, nonnegative function on $[a, b]$. Then the surface area S generated by revolving the portion of the curve $y = f(x)$ between $x = a$ and $x = b$ about the x -axis is

$$S = 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} dx$$

5) EXAMPLE:

Find the surface area of the portion of the sphere generated by revolving the curve $y = \sqrt{1-x^2}$, $-1 \leq x \leq 1$ about the x -axis.

Solution:

$$\text{Since } f(x) = \sqrt{1-x^2}, \quad f'(x) = \frac{-x}{\sqrt{1-x^2}}$$

$$1 + [f'(x)]^2 = 1 + \frac{x^2}{1-x^2} = \frac{1-x^2+x^2}{1-x^2} = \frac{1}{1-x^2}$$

$$\sqrt{1 + [f'(x)]^2} = \frac{1}{\sqrt{1-x^2}}$$

Thus from surface area formula

$$S = 2\pi \int_{-1}^1 \sqrt{1-x^2} \cdot \frac{1}{\sqrt{1-x^2}} dx = 2\pi \int_{-1}^1 1 dx = 2\pi [x]_{-1}^1$$

$$= 2\pi (1 - (-1)) = 4\pi$$

$$S = 4\pi$$

Page 1

Lecture # 38

Work and Definite Integral

In this lecture we will discuss

- * Work done by a constant force
- * Work done by a variable force
- * Fluid Pressure
- * Pascal's Principle

Work done by a constant force

If an object moves a distance d along a line while a CONSTANT force F is acting on it, then work W done on the object by the force F is defined as

$$W = F \cdot d \Rightarrow \text{Work} = \text{Force} \times \text{distance}$$

- Distance unit is meter represented by m
- Force units are Pounds (lbs), Dynes or Newton (N).

One Dyne = the force needed to give a mass of 1 gram an acceleration of 1 cm/s^2
 1 Newton = the force needed to give a mass of 1 Kg an acceleration of 1 m/s^2

Example

An object moves 5 ft along a line while subjected to a force of 100 lbs in its direction of motion. The work done is

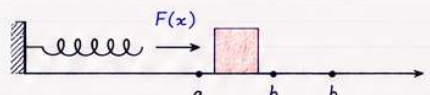
$$W = F \cdot d = 100 (5) = 500 \text{ ft. lbs.}$$

Work done by a variable force

- So far the force was a constant force.
- What if it's changing constantly?
- The equation of Work we just saw will not work.
- We need Calculus

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In this figure, we have a block subjected to the force of a compressed spring. As the block moves from a to b , the spring gets un-compressed and the force it applied diminishes. So here is a case where the force varies with the position x of the spring. So the force $F(x)$ is a function of the position of the block on the x -axis.

We need to define work done by a variable force. This, in turn, will give us the answer to calculate it as well through definite integral.

As before we subdivide the interval $[a, b]$ into subintervals with coordinates

$$a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b$$

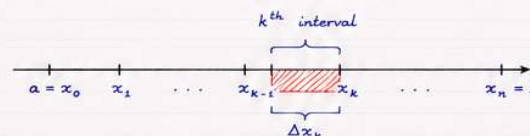
and widths $\Delta x_1, \Delta x_2, \dots, \Delta x_n$ etc.

Work and definite integral

Let's consider the k^{th} interval $[x_{k-1}, x_k]$ and the force F over it.

If the interval is small, then the force F will be almost constant on it.

We can approx F on this interval by $F(x_k^*)$ where x_k^* is a point in the k^{th} interval. The width of this interval will be $\Delta x_k = x_k - x_{k-1}$.



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So the work done on this interval is

$$W_k \approx F(x_k^*) \Delta x_k$$

So the work done over the whole interval is

$$W \approx W_1 + W_2 + \dots + W_n = \sum_{k=1}^n F(x_k^*) \Delta x_k$$

Let the largest subinterval go to 0 to get

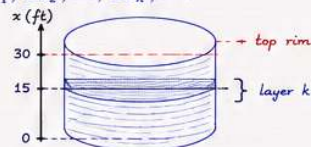
$$W = \lim_{\max \Delta x_k \rightarrow 0} \sum_{k=1}^n F(x_k^*) \Delta x_k = \int_a^b F(x) dx$$

Example

A cylindrical water tank of radius 10 ft and height 30 ft is half filled with water. How much force is needed to pump all the water over the upper rim of the tank?

Solution

Introduce a coordinate line as shown in figure, imagine the water to be divided into n thin layers with thicknesses $\Delta x_1, \Delta x_2, \dots, \Delta x_n, \dots$



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How much work force is required to move the k^{th} layer of water above the rim? The force required to move the k^{th} layer equals the weight of the layer, which can be found by multiplying its volume by weight density of water.

Volume of layer k
 = (area of base) $\times$ (thickness)
 = $A \Delta x_k$

where $A = \pi r^2 = \pi (10)^2 = 100\pi \text{ (ft}^2\text{)}$

Weight density of water
 $p = 62.4 \text{ lb/ft}^3$

Weight of layer k
 = $p \times$ (volume of layer k)
 = $62.4 \times (100\pi) \Delta x_k$

Force to lift this layer

$$F_k = 62.4 (100\pi) \Delta x_k$$

This layer is at a depth x_k from the bottom, so it must be lifted a distance $(30 - x_k)$ feet to reach the top rim.

Work for layer k

$$W_k = F_k (30 - x_k) = 62.4 (100\pi) (30 - x_k) \Delta x_k$$

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Total work to pump all the water out

$$\begin{aligned} W &= \lim_{\max \Delta x_k \rightarrow 0} \sum_{k=1}^n W_k \\ &= \lim_{\max \Delta x_k \rightarrow 0} \sum_{k=1}^n 62.4 (100\pi) (30 - x_k) \Delta x_k \\ &= 62.4 (100\pi) \lim_{\max \Delta x_k \rightarrow 0} \sum_{k=1}^n (30 - x_k) \Delta x_k \\ &= 62.4 (100\pi) \int_0^{15} (30 - x) dx \end{aligned}$$

Evaluate:

$$\begin{aligned} W &= 6240\pi \left[30x - \frac{x^2}{2} \right]_0^{15} \\ &= 6240\pi \left[30(15) - \frac{15^2}{2} - 0 \right] \\ &= 6240\pi (450 - 112.5) \\ &= 6240\pi (337.5) \\ &= 2,106,000\pi \text{ ft. lb} \\ &\approx 6,616,392 \text{ ft. lb} \end{aligned}$$

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Pressure is defined as Force per unit Area.

$$P = \frac{F}{A} = ph$$

Where p is weight density and h is depth below surface of fluid.

Fluid Pressure

If a flat surface of area A is submerged horizontally in a fluid at a depth h , then the fluid exerts a force F perpendicular to the surface.

$$F = p h A$$

p is the weight density of the fluid.
 p for water is 62.4 lb/ft^3

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It is a fact from Physics that the shape of the container containing the fluid does not in any way effect F .

If the three containers here have the same area for their bases and the fluid has the same height, then the F on the bases will be equal.



Figure 6.8.1

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Pascal's Principle

Fluid pressure is the same in all directions at a given height.

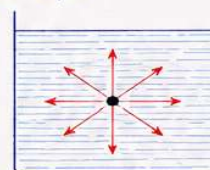


Figure 6.8.2

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Lecture #39: IMPROPER INTEGRAL

In this lecture we will study

- ★ Integrals over Infinite Interval
- ★ Integrals whose Integrands become Infinite

1) INTEGRALS OVER INFINITE INTERVAL

As we saw before, for a given continuous function f , the definite integral is

$$\int_a^b f(x) dx \quad (\text{finite interval})$$

It is assumed that the interval $[a, b]$ is finite.

What if we look at $[a, +\infty)$ and the corresponding integral

$$\int_a^{+\infty} f(x) dx$$

In this case, we define what is called an improper integral over an infinite interval.

★ What does it mean to integrate all the way to $+\infty$?

The answer will be clear if we define this integral as a limit in the following way.

$$\int_a^{+\infty} f(x) dx = \lim_{l \rightarrow +\infty} \int_a^l f(x) dx$$

What this does is to first turn the integral into more familiar form of over a finite interval, and then we let the upper limits of the interval approach ∞ and see what happens to the answer we had got earlier.

★ If this limit exists, then we say that the Improper Integral **Converges**, and the value of the limit is assigned to the integral.

★ If the limit does not exist, then we say that the Improper Integral **Diverges**, and no finite value is assigned.

IMPORTANT NOTE

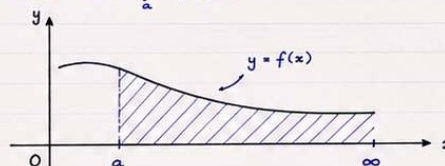
To check convergence of $\int_a^{+\infty} f(x) dx$

- 1) Evaluate $\int_a^l f(x) dx$
- 2) Take $\lim_{l \rightarrow +\infty}$ of the result.
- 3) If limit is finite $\Rightarrow$ **Convergent**
If limit is $\pm\infty$ or does not exist $\Rightarrow$ **Divergent**.

Improper integrals arise when the interval of integration is infinite or the integrand becomes infinite.

GEOMETRIC IDEA:

Consider $y = f(x) \geq 0$ on $[a, \infty)$. The area under the curve from a to ∞ is $\int_a^{+\infty} f(x) dx$.

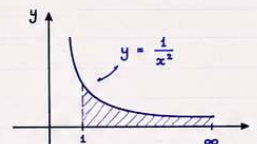


We approximate by taking area up to $x = l$ and then let $l \rightarrow \infty$.

EXAMPLES:

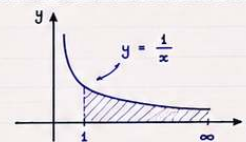
1) $f(x) = \frac{1}{x^2}$, $a = 1$

$$\begin{aligned} \int_1^{+\infty} \frac{1}{x^2} dx &= \lim_{l \rightarrow +\infty} \int_1^l x^{-2} dx \\ &= \lim_{l \rightarrow +\infty} \left[-\frac{1}{x} \right]_1^l \\ &= \lim_{l \rightarrow +\infty} \left(-\frac{1}{l} + 1 \right) = 1 \quad (\text{Converges}) \end{aligned}$$



2) $f(x) = \frac{1}{x}$, $a = 1$

$$\begin{aligned} \int_1^{+\infty} \frac{1}{x} dx &= \lim_{l \rightarrow +\infty} \int_1^l \frac{1}{x} dx \\ &= \lim_{l \rightarrow +\infty} [\ln x]_1^l \\ &= \lim_{l \rightarrow +\infty} (\ln l - \ln 1) = \lim_{l \rightarrow +\infty} \ln l = +\infty \quad (\text{Diverges}) \end{aligned}$$



COMPARISON:

$y = \frac{1}{x^2}$	$y = \frac{1}{x}$
- Approaches $y = 0$ much faster. - Area $\int_1^{+\infty} \frac{1}{x^2} dx$ is finite (Convergent) - Volume when rotated about x-axis is finite.	- Approaches $y = 0$ slowly. - Area $\int_1^{+\infty} \frac{1}{x} dx$ is infinite (Divergent) - Volume when rotated about x-axis is infinite.

Since $\frac{1}{2} > \frac{1}{4}$, $\frac{1}{8} > \frac{1}{64}$, etc.

so $\frac{1}{x^2}$ is much smaller than $\frac{1}{x}$ for large x .

Lecture #39: IMPROPER INTEGRAL

(Part - 2)

We would like to find the volume of this solid.

The cross section is a disk with radius $f(x)$.

For $f(x) = \frac{1}{x}$, we get for volume

$$V = \pi \lim_{l \rightarrow +\infty} \int_1^l \left(\frac{1}{x}\right)^2 dx = \pi \lim_{l \rightarrow +\infty} \int_1^l \frac{1}{x^2} dx = \pi$$

So we can find out how much paint can be held in this solid, but we cannot paint the inside of the solid !!!

We can also have an improper integral of this type:

$$\lim_{l \rightarrow -\infty} \int_1^b f(x) dx = \lim_{l \rightarrow -\infty} \int_l^b f(x) dx$$

② INTEGRALS OVER INFINITE INTERVAL (Type - II)

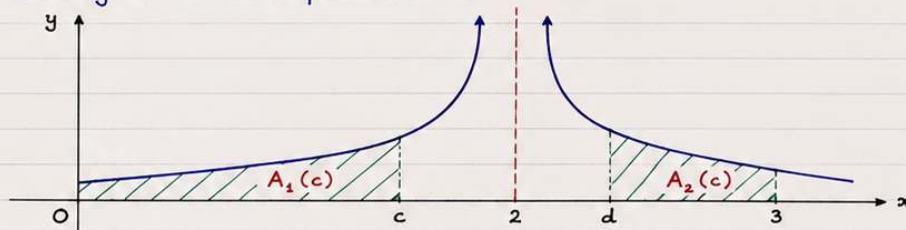
We saw earlier that if a function f is not bounded on an interval $[a, b]$, then f is not integrable on $[a, b]$.

The integral $\int_0^3 \frac{1}{(x-2)^{2/3}} dx$ is unbounded at $x=2$ in $[0, 3]$.

We can get around this problem by doing the following:

Break the interval $[0, 3]$ at the point $x=2$ where f is unbounded and integrate over the two subintervals $[0, c]$ and $[c, 3]$ where c is any number such that $0 < c < 2 < 3$. Then we let $c \rightarrow 2^-$ and $c \rightarrow 2^+$ and see what happens to the integrals.

Geometrically it can be represented as:



If f is continuous on $[a, b]$ except that at some point c such that $a < c < b$, $f(x)$ becomes infinite as x goes to c from left or right.

If both improper integrals $\int_a^c f(x) dx$ and $\int_c^b f(x) dx$ converge then we say that the improper integral $\int_a^b f(x) dx$ converges as we define:

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

Lecture # 40: L'HOPITAL'S RULE AND INDETERMINATE FORMS

- ★ L'Hopital's Rule and 0/0
- ★ Indeterminate form of type ∞/∞
- ★ Indeterminate form of type $0 \cdot \infty$
- ★ Indeterminate forms of type 0^0 , ∞^0 and 1^∞ and $\infty - \infty$

If we look at

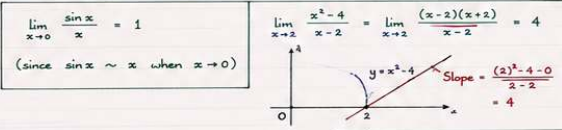
$$\lim_{x \rightarrow 0} \frac{\sin x}{x} \quad \text{and} \quad \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

The top and the bottom both approach 0 and so these types of limits are called **indeterminate forms of type 0/0**.

This kind of limit can converge in which case it will have finite real values, or it can diverge.

The value, if it converges, is not obvious right away.

The first expression can be factored and bottom cancelled to get a finite value and the second expression requires geometrical observation as shown below.

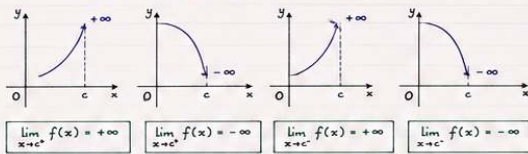


We need a more general concept that work almost always.
L'Hopital's Rule provide us solution that we are looking for.

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SOME NOTATION

In the four parts of figure below,



When we want to indicate that one of these four situations occurs without specifying which one, we shall write

$$\lim_{x \rightarrow c} f(x) = \pm \infty \quad (\text{i.e. tends to infinity})$$

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If $\lim_{x \rightarrow c} f(x) = 0$ and $\lim_{x \rightarrow c} g(x) = \infty$ (or $-\infty$),

then $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = 0$ which is not of type $\frac{\infty}{\infty}$ or $\frac{0}{0}$.

But it can be converted.

If $\lim_{x \rightarrow c} f(x) = 0$ and $\lim_{x \rightarrow c} g(x) = \infty$, then

$$\lim_{x \rightarrow c} \left(\frac{1}{g(x)} \right) = 0$$

Limit problems of this type can be converted to the form of 0/0 by writing

$$f(x)g(x) = \frac{f(x)}{\frac{1}{g(x)}} \quad (\text{Now both numerator and denominator approach } 0)$$

INDETERMINATE FORMS OF TYPE 0^0 , ∞^0 AND 1^∞ AND $\infty - \infty$

Limits of the form 0^0 , ∞^0 , 1^∞ and $\infty - \infty$ give rise to indeterminate form of type 0^0 , ∞^0 and 1^∞ .

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HOW WE CAN USE THIS THEOREM, HERE ARE SOME STEPS TO BE FOLLOWED

L'Hopital's Rule (0/0 form)

Let f and g be functions such that

(i) $\lim_{x \rightarrow c} f(x) = 0$ and $\lim_{x \rightarrow c} g(x) = 0$
(or both $\rightarrow 0$ as $x \rightarrow \pm \infty$)

(ii) f and g are differentiable near c (except possibly at c)

(iii) $g'(x) \neq 0$ near c .

Then,

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

provided the limit on the right exists (or is $\pm \infty$).

Steps to apply L'Hopital's Rule :

- 1 Check whether the limit is of type $\frac{0}{0}$.
- 2 Differentiate the numerator and denominator.
- 3 Form the new limit $\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$.
- 4 Evaluate the limit.
- 5 If the new limit is still of type $\frac{0}{0}$, repeat the process (differentiate again).
- 6 Stop when you get a limit that is not of type $\frac{0}{0}$.

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INDETERMINATE FORM OF TYPE ∞/∞

An indeterminate form of type $\frac{\infty}{\infty}$ is a limit i.e. $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$ in which

$\lim_{x \rightarrow c} f(x) = \infty$ and $\lim_{x \rightarrow c} g(x) = \infty$. Some examples are

$$\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x - 3}$$

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{x}$$

$$\lim_{x \rightarrow \infty} \frac{e^x}{\sqrt{x}}$$

The following version of L'Hopital Rule is used for problems like this

L'Hopital's Rule (∞/∞ form)

If $\lim_{x \rightarrow c} f(x) = \infty$ and $\lim_{x \rightarrow c} g(x) = \infty$ (or both $-\infty$)

and if f and g are differentiable near c with $g'(x) \neq 0$ near c , then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

provided the limit on the right exists (or is $\pm \infty$).

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All these types are treated by first introducing a dependent variable

$$(i) \ y = f(x) > 0 \Rightarrow x = \ln y$$

Now, as $x \rightarrow c$, $y = f(x) \rightarrow 0^+$ or ∞ and we can apply L'Hopital's Rule.

$$(ii) \ y = f(x) > 0 \Rightarrow x = \ln y$$

Now, as $x \rightarrow c$, $y = f(x) \rightarrow \infty$ and we can apply L'Hopital's Rule.

$$(iii) \ y = f(x) > 0 \Rightarrow x = \frac{\ln y}{g(x)} \text{ (if } g(x) \rightarrow 0 \text{)}$$

Now, as $x \rightarrow c$, $y^{g(x)} = e^{g(x) \ln y}$ becomes the form 1^∞ (or 0^0) and we apply L'Hopital's Rule.

$$(iv) \ \infty - \infty \text{ form} \Rightarrow \text{Algebraic simplification}$$

Simplify the expression if possible. If it reduces to $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then apply L'Hopital's Rule.

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Lecture #41:SEQUENCES AND MONOTONE SEQUENCES

- ★ Definition of a sequence
- ★ Graphs of sequences
- ★ Limit of a sequence
- ★ Recursive sequence
- ★ Testing for monotonicity
- ★ Eventually monotonic sequences
- ★ Convergence of monotonic sequence
- ★ More on convergence intuitively

A sequence is simply an ordered list of numbers written in a definite order.

😊

1) DEFINITION OF A SEQUENCE

- A sequence in math is a succession of numbers e.g. 2, 4, 6, 8, ... & 1, 2, 3, 4, ...
- The numbers in a sequence are called terms of a sequence.
- Each term has a positional name, like 1st term, 2nd term etc and we can write them as a_1, a_2 etc.
- It is convenient to write a sequence as a formula.

2, 4, 6, 8, ... can be written as a formula as $\{2n\}_{n=1}^{+\infty}$

We will write a sequences like $a_1, a_2, a_3, \dots$ as

$$\{a_n\}_{n=1}^{+\infty}$$

Here is a formal definition of a sequence

Definition:

A sequence is a function whose domain is the set of positive integers $N = \{1, 2, 3, \dots\}$.

We usually denote the n^{th} term of the sequence by a_n or u_n .

Thus a sequence = $\{a_n\}_{n=1}^{+\infty}$

where $a_n : N \rightarrow R$

Note:

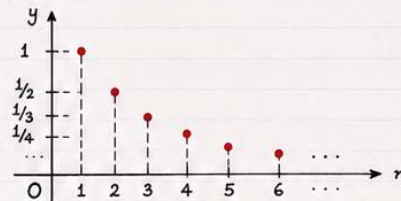
The index n tells us the position of the term in the sequence.

For example, a_1 is the first term, a_2 is the second term and so on.

2) GRAPHS OF SEQUENCES

As we see that sequences are functions, we can talk about the graphs of them.

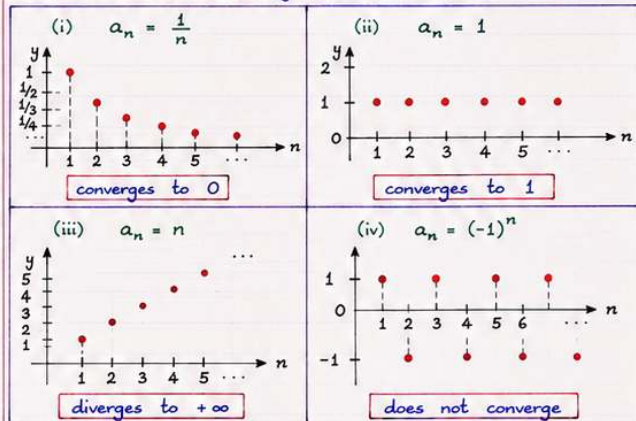
The graph of the sequence $\left\{\frac{1}{n}\right\}_{n=1}^{+\infty}$ is the graph of the function equation $y = \frac{1}{n}$ for $n = 1, 2, 3, \dots$



In sequence graphs, we plot discrete points only at integer values of n .

3) LIMIT OF A SEQUENCE

Here are graphs of four sequences, each of which behaves differently as n increases.

4) RECURSIVE SEQUENCE

Some sequences are defined by specifying one or more initial terms and giving a formula that relates each subsequent term to the term that precedes it, such sequences are said to be defined recursively.

Examples:

- $a_1 = 1, a_n = a_{n-1} + 2$ for $n \geq 2$
- $a_1 = 1, a_2 = 1, a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$

Each term is defined in terms of previous term(s).

Lecture #41: SEQUENCES AND MONOTONE SEQUENCES (Part-2)

5) MONOTONICITY AND TESTING FOR MONOTONICITY

- Let $\{a_n\}$ be a sequence.

DEFINITION: MONOTONE SEQUENCE

- (i) Increasing (Monotone Increasing):

The sequence $\{a_n\}$ is increasing if

$$a_1 \leq a_2 \leq a_3 \leq \dots \leq a_n \leq \dots$$

i.e. $a_{n+1} \geq a_n$ for all n .

- (ii) Decreasing (Monotone Decreasing):

The sequence $\{a_n\}$ is decreasing if

$$a_1 \geq a_2 \geq a_3 \geq \dots \geq a_n \geq \dots$$

i.e. $a_{n+1} \leq a_n$ for all n .

- (iii) Monotone: A sequence is monotone if it is either increasing or decreasing.

TESTING FOR MONOTONICITY

To test whether $\{a_n\}$ is increasing or decreasing, we can compare a_{n+1} and a_n .

- If $a_{n+1} \geq a_n$ for all n , then $\{a_n\}$ is increasing.
- If $a_{n+1} \leq a_n$ for all n , then $\{a_n\}$ is decreasing.
- If the inequality is strict ($>$ or $<$) for all n , then the sequence is strictly increasing or strictly decreasing.

- ★ Monotone Sequences (general) may oscillate around a value or not approach any particular value. But monotone sequences with positive terms have special convergence properties.

6) MONOTONE SEQUENCES WITH POSITIVE TERMS

Assume
 $a_n > 0$
for all n .

- (i) If $\{a_n\}$ is increasing and bounded above, then it converges.
(ii) If $\{a_n\}$ is decreasing and bounded below, then it converges.

7) TESTING FOR MONOTONICITY USING DERIVATIVES

If $f(n) = a_n$ is the n^{th} term of a sequence and f is differentiable for $x \geq 1$, then we have the following results:

(i) If $f'(x) \geq 0$ for $x \geq 1$, then $\{a_n\}$ is increasing.	<u>Reason:</u> $f'(x) \geq 0 \Rightarrow f$ is increasing on $[1, \infty) \Rightarrow a_{n+1} = f(n+1) \geq f(n) = a_n$.
(ii) If $f'(x) \leq 0$ for $x \geq 1$, then $\{a_n\}$ is decreasing.	<u>Reason:</u> $f'(x) \leq 0 \Rightarrow f$ is decreasing on $[1, \infty) \Rightarrow a_{n+1} = f(n+1) \leq f(n) = a_n$.
(iii) If $f'(x) = 0$ for $x \geq 1$, then $\{a_n\}$ is constant.	<u>Reason:</u> $f'(x) = 0 \Rightarrow f(x) = \text{constant} \Rightarrow a_n = \text{constant}$.

8) EVENTUALLY MONOTONIC SEQUENCES

A sequence $\{a_n\}$ is eventually monotone if there exists an integer N such that the sequence is monotone for all $n \geq N$.

In other words, after some point, the sequence becomes monotone.

9) CONVERGENCE OF MONOTONIC SEQUENCES

- (i) Every bounded increasing sequence converges.
(ii) Every bounded decreasing sequence converges.

So if $\{a_n\}$ is monotone and bounded, then it must have a finite limit.

10) AN INTUITIVE VIEW OF CONVERGENCE

Informally stated, the convergence or divergence of a sequence does not depend on the behavior of the "initial terms" of the sequence, but rather on the behavior of the "tail end". Thus for $\{a_n\}$ to converge to a limit L , it does not matter if the initial terms are far from L , just so the terms in the sequence are eventually arbitrarily close to L . This being the case, one can add, delete, or alter finitely many terms without affecting the convergence, divergence, or the limit (if it exists).

Key Point

Convergence depends on the tail, not on the first few terms.



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Lecture # 42

Infinite Series

Definition 11.3.1

An Infinite series is an expansion that can be written in the form

$$\sum_{k=1}^{\infty} u_k = u_1 + u_2 + \dots + u_k + \dots$$

The numbers $u_1, u_2, \dots$ are called the terms of the series.

What does it mean to add infinitely many terms?

Can we add $1+2+3+4+\dots$ All the way to infinity?

This is physically and mentally impossible!

But we can compute an infinite sum using the idea of limits.

Look at this decimal expansion

$$0.333333 \dots$$

We can rewrite this as an infinite series

$$0.3 + 0.03 + 0.003 + 0.0003 + \dots$$

Or as

$$\frac{3}{10} + \frac{3}{100} + \frac{3}{1000} + \dots = \sum_{k=1}^{\infty} \frac{3}{10^k}$$

Note that we already know that $0.333 \dots$ is really $\frac{1}{3}$.

So our sum of the above series should also be $\frac{1}{3}$.

Let's work out a definition for the sum of an infinite series.

Consider the following sequences of (finite) sums

$$S_1 = 0.3 = \frac{3}{10}$$

$$S_2 = 0.3 + 0.03 = \frac{3}{10} + \frac{3}{100} = 0.33$$

$$S_3 = 0.3 + 0.03 + 0.003 = \frac{3}{10} + \frac{3}{100} + \frac{3}{1000} = 0.333$$

The sequence of numbers $s_1, s_2, \dots$ can be viewed as a succession of approximations to the "sum" of the infinite series. We know that the answer is $\frac{1}{3}$, so it is better to say that the approximations are close to this at the start, and get closer and closer to this number as more and more terms are added together. This suggests that if we take the limit of the sequence of approximations, we should get $\frac{1}{3}$.

Note that the n^{th} term of the approximating series is

$$s_n = \frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \dots + \frac{3}{10^n}$$

Taking its limit gives (limit of the n^{th} partial sum of the series)

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$$\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left(\frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \dots + \frac{3}{10^n} \right)$$

Now we do some algebraic manipulation here

$$\lim_{n \rightarrow \infty} s_n = S = \frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \dots \quad (1)$$

Multiply both sides by $\frac{1}{10}$

$$\frac{1}{10} S = \frac{3}{10^2} + \frac{3}{10^3} + \frac{3}{10^4} + \dots \quad (2)$$

Subtract (2) from (1)

$$S - \frac{1}{10} S = \frac{3}{10}$$

$$\left(1 - \frac{1}{10} \right) S = \frac{3}{10}$$

$$\frac{9}{10} S = \frac{3}{10} \Rightarrow S = \frac{1}{3}$$

Now we can formalize the idea of a sum of an infinite series

DEFINITION 11.3.2 (Sum of an Infinite Series)

Let $\sum_{k=1}^{\infty} u_k = u_1 + u_2 + u_3 + \dots$ be

an infinite series. Let

$$s_n = u_1 + u_2 + u_3 + \dots + u_n = \sum_{k=1}^n u_k$$

be the sequence of partial sums.

If $\lim_{n \rightarrow \infty} s_n = S$ exists (is finite),

then the series is said to converge

and S is called the sum of the series.

$$\text{We write } \sum_{k=1}^{\infty} u_k = S$$

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Geometric Series

A geometric series is one of the form

$$a + ar + ar^2 + ar^3 + \dots + ar^{k-1} + \dots$$

Each term is obtained by multiplying the previous one by a constant number, r , and this number is called the ratio for the series.

Examples are

$$\bullet 1 + 2 + 4 + 8 + \dots \quad (\text{here } a=1, r=2)$$

$$\bullet 1 + 1 + 1 + 1 + \dots \quad (\text{here } a=1, r=1)$$

Harmonic Series

A Harmonic series is of the type

$$\sum_{k=1}^{\infty} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

It may look like this series converges as each successive term is smaller than the first, but in fact this series diverges!

Convergence Tests

Here are some tests that are used to determine if a series converges or diverges.

These tests are applied on the k^{th} term of a series (general term), and not on the n^{th} partial sum as we did earlier to derive the definition of the sum of an infinite series.

This theorem tells us when a series diverges

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THEOREM 11.4.1 (Divergence Test)

Let $\sum_{k=1}^{\infty} u_k$ be a series.

(a) If $\lim_{k \rightarrow \infty} u_k \neq 0$ or does not exist, then the series diverges.

(b) If $u_k > 0$ for all k and $\lim_{k \rightarrow \infty} u_k = 0$, this test is inconclusive.

The alternative form of part (a) is sufficiently important that we state it separately

THEOREM 11.4.2

If $\lim_{k \rightarrow \infty} u_k \neq 0$, then $\sum_{k=1}^{\infty} u_k$ diverges.

Example

The series $\sum_{k=1}^{\infty} \frac{1}{k+1} = 1 + \frac{1}{2} + \frac{1}{3} + \dots$

Diverges: since

$$\lim_{k \rightarrow \infty} \frac{1}{k+1} = \lim_{k \rightarrow \infty} \frac{1}{k} = 0 \neq 1$$

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Algebraic Properties of Infinite Series

Let $\sum_{k=1}^{\infty} u_k$ and $\sum_{k=1}^{\infty} v_k$ be convergent series.

$$1. \quad c \sum_{k=1}^{\infty} u_k = \sum_{k=1}^{\infty} cu_k \quad (c \text{ is any constant})$$

$$2. \quad \sum_{k=1}^{\infty} (u_k + v_k) = \sum_{k=1}^{\infty} u_k + \sum_{k=1}^{\infty} v_k$$

$$3. \quad \sum_{k=1}^{\infty} (u_k - v_k) = \sum_{k=1}^{\infty} u_k - \sum_{k=1}^{\infty} v_k$$

Example

Find the sum of the series $\sum_{k=1}^{\infty} \left(\frac{1}{3^k} - \frac{2}{5^k} \right)$

Solution: The series

$$\sum_{k=1}^{\infty} \frac{1}{3^k} = \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$$

is a convergent series ($a = \frac{1}{3}$, $r = \frac{1}{3}$)

and the series

$$\sum_{k=1}^{\infty} \frac{2}{5^k} = \frac{2}{5} + \frac{2}{25} + \frac{2}{125} + \dots$$

is also a convergent series ($a = \frac{2}{5}$, $r = \frac{1}{5}$).

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Thus, from Theorem 11.4.3(a) and 11.3.3 the given series converges and

$$\begin{aligned} \sum_{k=1}^{\infty} \left(\frac{1}{3^k} - \frac{2}{5^k} \right) &= \sum_{k=1}^{\infty} \frac{1}{3^k} - \sum_{k=1}^{\infty} \frac{2}{5^k} \\ &= \frac{1}{1-\frac{1}{3}} - \frac{2}{1-\frac{1}{5}} = \frac{1}{\frac{2}{3}} - \frac{2}{\frac{4}{5}} = \frac{1}{2} - \frac{2}{4} = \frac{1}{2} - \frac{1}{2} = 0 \end{aligned}$$

THEOREM 11.4.4 (The Integral Test)

Let $\sum_{k=1}^{\infty} u_k$ be a series with positive terms, and let $f(x)$ be the function that results when k is replaced by x in the formula for u_k .

If f is decreasing and continuous on the interval $[a, +\infty)$, then

$$\sum_{k=1}^{\infty} u_k \quad \text{and} \quad \int_a^{\infty} f(x) dx$$

both converge or both diverge.

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This theorem allows us to study the convergence of a series by studying a related integral.

This integral is the improper integral over the interval $[1, +\infty)$ and the function is the k^{th} term of the series.

It is interesting that the integral represents a continuous phenomenon, while the summation a discrete one!

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p-Series

p -series or hyper-harmonic series is an infinite series of the form $\sum_{k=1}^{\infty} \frac{1}{k^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \dots$

The following theorem tells us whether a p -series converges or diverges

THEOREM 11.4.5 (p-Series Test)

The p -series $\sum_{k=1}^{\infty} \frac{1}{k^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

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Examples

$$1) \quad \sum_{k=1}^{\infty} \frac{1}{k^2} = 1 + \frac{1}{4} + \frac{1}{9} + \dots \quad \text{Here } p = 2 > 1$$

$\Rightarrow$ converges.

$$2) \quad \sum_{k=1}^{\infty} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \dots \quad \text{Here } p = 1$$

$\Rightarrow$ diverges.

$$3) \quad \sum_{k=1}^{\infty} \frac{1}{\sqrt{k}} = 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots \quad \text{Here } p = \frac{1}{2} < 1$$

$\Rightarrow$ diverges.

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Lecture #43: ADDITIONAL CONVERGENCE TESTS

In this section we shall develop some additional convergence tests for series with positive terms.

THEOREM 11.5.1 (The Comparison Test)

Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be series with nonnegative terms and suppose that

$$a_1 \leq b_1, a_2 \leq b_2, a_3 \leq b_3, \dots, a_k \leq b_k, \dots$$

- If the "bigger series" $\sum_{k=1}^{\infty} b_k$ converges then the "smaller series" $\sum_{k=1}^{\infty} a_k$ also converges.
- If the "smaller series" $\sum_{k=1}^{\infty} a_k$ diverges, then the "bigger series" $\sum_{k=1}^{\infty} b_k$ also diverges.

- ★ This test basically compares two series with each other, and if the series in which each term is bigger than that of the other series converges, then so does the smaller one.
- ★ If the series in which each term is smaller than that of the other series diverges, then so does the bigger one.

We will see more of this later. For now, we shall use the comparison test to develop some other tests that are easier to apply.

THEOREM 11.5.2 (The Ratio Test)

Let $\sum_{k=1}^{\infty} u_k$ be a series with positive terms and suppose that $\rho = \lim_{k \rightarrow \infty} \frac{u_{k+1}}{u_k}$,

- If $\rho < 1$, the series converges.
- If $\rho > 1$ or $\rho = +\infty$, the series diverges.
- If $\rho = 1$, the series may converge or diverge, so that another test must be tried.

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If $\lim_{k \rightarrow \infty} f(x) = 0$ and $\lim_{k \rightarrow \infty} g(x) = \infty$ (or $-\infty$), then $\lim_{k \rightarrow \infty} f(x)g(x) = 0 \cdot \infty$ which is not of type $\frac{\infty}{\infty}$ or $\frac{0}{0}$. But it can be converted.

If $\lim_{k \rightarrow \infty} f(x) = 0$ and $\lim_{k \rightarrow \infty} g(x) = \infty$, then

$$\lim_{k \rightarrow \infty} \left(\frac{f(x)}{g(x)} \right) = 0$$

Limit problems of this type can be converted to the form of $0/0$ by writing

$$f(x)g(x) = \frac{f(x)}{1/g(x)} \quad (\text{Now both numerator and denominator approach } 0)$$

INDETERMINATE FORMS OF TYPE 0^0 , ∞^0 AND 1^∞ AND $\infty - \infty$

Limits of the form 0^0 , ∞^0 , 1^∞ and $\infty - \infty$ give rise to indeterminate form of type 0^0 , ∞^0 and 1^∞ .

All these types are treated by first introducing a dependent variable

The Comparison Test

We shall now discuss some techniques for applying the comparison test (Theorem 11.5.1).

There are two basic steps required to apply the comparison test to a series $\sum_{k=1}^{\infty} u_k$ of positive terms:

- Guess whether the series $\sum_{k=1}^{\infty} u_k$ converges or diverges.
- Find a series that proves the guess to be correct. Thus if the guess is divergence, we must find a divergent series whose terms are "smaller" than the corresponding terms of $\sum_{k=1}^{\infty} u_k$, and if the guess is convergence, we must find a convergent series whose terms are "bigger" than the corresponding terms of $\sum_{k=1}^{\infty} u_k$.

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Sometimes the following result is easier to apply than ratio test.

THEOREM 11.5.3 (The Root Test)

Let $\sum_{k=1}^{\infty} u_k$ be a series with positive terms and suppose that $\rho = \lim_{k \rightarrow \infty} \sqrt[k]{u_k}$.

- If $\rho < 1$, the series converges.
- If $\rho > 1$ or $\rho = +\infty$, the series diverges.
- If $\rho = 1$, the series may converge or diverge, so that another test must be tried.

Example: $\sum_{k=2}^{\infty} \left(\frac{2k-1}{2k+1} \right)^{5/4}$

Use the root test to determine whether the following series converge or diverge.

Solution: The series diverges, since

$$\rho = \lim_{k \rightarrow \infty} \sqrt[k]{\left(\frac{2k-1}{2k+1} \right)^{5/4}} = \lim_{k \rightarrow \infty} \frac{2k-1}{2k+1} = 1 > 1$$

We have not really talked about the comparison test so far. To apply comparison test, we use an alternative version of this test that is easier to work with.

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INFORMAL PRINCIPLE 11.6.1

Constant terms in the denominator of u_k can usually be deleted without affecting the convergence or divergence of the series.

Example: Use the above informal principle to help guess whether the following series converge or diverge.

$$\sum_{k=1}^{\infty} \frac{1}{2k+1}$$

Solution: Deleting the constant 1 suggests that

$$\frac{1}{2k+1} \text{ behaves like } \frac{1}{2k} \quad \& \quad \sum_{k=1}^{\infty} \frac{1}{2k} = \sum_{k=1}^{\infty} \frac{1}{2} \cdot \frac{1}{k}$$

The modified series is a convergent geometric series, so the given series is likely to converge.

INFORMAL PRINCIPLE 11.6.2

If a polynomial in k appears as a factor in the numerator or denominator of u_k , all but the highest power of k in the polynomial may usually be deleted without affecting the convergence or divergence of the series.

Example: Use the above principle to help guess whether the following series $\sum_{k=1}^{\infty} \frac{3k+1}{k^2+2k}$ converge or diverge.

Solution: Deleting the term $2k$ suggests that

$$\frac{3k+1}{k^2+2k} \text{ behaves like } \frac{3k}{k^2} = \frac{3}{k} \quad \& \quad \sum_{k=1}^{\infty} \frac{3}{k} = \frac{3}{k^{3/2}}$$

Since the modified series is a convergent p -series ($p=3/2$), the given series is likely to converge.

The Limit Comparison Test

The following result can be used to establish convergence or divergence by examining the limit of the ratio of the general term of the series in question with general term of a series whose convergence properties are known.

THEOREM 11.6.3 (The Limit Comparison Test)

Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be series with positive terms and suppose that $\rho = \lim_{k \rightarrow \infty} \frac{a_k}{b_k}$.

If ρ is finite and $\rho > 0$, then the series both converge or both diverge.

Example: Use the limit comparison test to determine whether the following series $\sum_{k=2}^{\infty} \frac{1}{k^2-2k}$ converge or diverge.

Example: Use the comparison test to determine whether the following series $\sum_{k=1}^{\infty} \frac{1}{k-1/4}$ converge or diverge.

Solution: As $\frac{1}{k-1/4} > \frac{1}{k}$ for $k \geq 1$. Note that the series behaves like the divergent harmonic series, and hence is likely to diverge. Thus our goal is to find a divergent series that is "smaller" than the given series. We can do this by dropping the constant $-1/4$ in the denominator, thereby decreasing the size of the general term:

$$\frac{1}{k-1/4} > \frac{1}{k} \quad \text{for } k = 1, 2, \dots$$

Thus the given series diverges by the comparison test, since $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges.

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Lecture # 44

Alternating Series ;
Conditional Convergence

- * So far we have seen infinite series that have positive terms only.
- * We have also defined the limit and the Sum of such infinite series.
- * Now we look at series that have terms which have alternating signs, known as Alternating series.

Example

1) $1 - 1 + 1 - 1 + \dots$

2) $1 + 2 - 3 + 4 - \dots$

More generally, an alternating series has one of the two following forms

$$\sum_{k=1}^{\infty} (-1)^{k+1} a_k = a_1 - a_2 + a_3 - a_4 + \dots \quad (1)$$

$$\sum_{k=1}^{\infty} (-1)^k a_k = -a_1 + a_2 - a_3 + a_4 - \dots \quad (2)$$

Note that all the terms a_k are to be taken as being positive.

The following theorem is the key result on convergence of alternating series.

(1)

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Theorem 11.7.1 (Alternating Series Test)

An alternating series of either form (1) or (2) converges if the following two conditions are satisfied:

(a) $a_1 > a_2 > a_3 > \dots > a_k > \dots$

(b) $\lim_{k \rightarrow \infty} a_k = 0$

This theorem tells us when an alternating series converges.

Example

Use the alternating series test to show that the following series converges

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

Solution :

We need to verify the two conditions.

(a) $a_k = \frac{1}{k}$

$$\frac{1}{1} > \frac{1}{2} > \frac{1}{3} > \dots > \frac{1}{k} > \dots \quad \checkmark$$

(b) $\lim_{k \rightarrow \infty} a_k = \lim_{k \rightarrow \infty} \frac{1}{k} = 0 \quad \checkmark$

Both conditions are satisfied. Hence by Theorem 11.7.1, the series converges.

Note : This series is called the Alternating Harmonic Series.

(2)

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Harmonic series diverges.

The alternating harmonic series converges.

Now we will look at the errors involved in approximating an alternating series with a partial sum.

Theorem 11.7.2

If an alternating series satisfies the conditions of the alternating series test, and if the sum S of the series is approximated by the n^{th} partial sum S_n , thereby resulting in an error of $S - S_n$, then

$$|S - S_n| < a_{n+1}$$

Moreover, the sign of error is the same as that of the coefficient of a_{n+1} in the series.

Example

The alternating series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + (-1)^{k+1} \frac{1}{k} + \dots$$

satisfies the condition of the alternating series test; hence the series has a sum S , which we know must lie between any two successive partial sums. In particular, it must lie between

(3)

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$$S_7 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} = \frac{319}{420}$$

and

$$S_8 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} = \frac{533}{840}$$

So

$$\frac{533}{840} < S < \frac{319}{420}$$

If we take $S \approx \ln 2$ then

$$\frac{533}{840} < \ln 2 < \frac{319}{420}$$

$$0.6345 < \ln 2 < 0.7596$$

The value of $\ln 2$, rounded to four decimal places, is 0.6931, which is consistent with these inequalities.

It follows from Theorem 11.7.2 that

$$|\ln 2 - S_7| = \left| \ln 2 - \frac{319}{420} \right| < a_8 = \frac{1}{8}$$

and

$$|\ln 2 - S_8| = \left| \ln 2 - \frac{533}{840} \right| < a_9 = \frac{1}{9}$$

(4)

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Interpretation :

- The true sum S always lies between two successive partial sums.
- The error in using S_n as an approximation is smaller than the next term a_{n+1} .
- The error has the same sign as the next term.

Summary

- * Alternating series: terms change signs.
- * Converges if:
 - (a) $a_1 > a_2 > a_3 > \dots$
 - (b) $\lim_{k \rightarrow \infty} a_k = 0$
- * Error estimate: $|S - S_n| < a_{n+1}$
- * Error sign = sign of next term.

(5)

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Important Points to Remember

- ① All a_k in alternating series test are positive.
- ② The conditions are sufficient, not necessary.
- ③ Partial sums of an alternating series oscillate around the sum S .
- ④ The approximation gets better as n increases because $a_{n+1} \rightarrow 0$.

Exam Tip

If you see $(-1)^{k+1}$ or $(-1)^k$ in the series, first check the two conditions of the alternating series test.

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Absolute and Conditional Convergence

Definition 11.7.3

A series $\sum_{k=1}^{\infty} u_k = u_1 + u_2 + u_3 + \dots$ is said to converge absolutely if the series of absolute values

$$\sum_{k=1}^{\infty} |u_k| = |u_1| + |u_2| + |u_3| + \dots$$

converges.

Example 1

The series $1 - \frac{1}{2} - \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} - \frac{1}{2^5} - \frac{1}{2^6} + \dots$ Converges absolutely since the series of absolute values

$$1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \frac{1}{2^5} + \frac{1}{2^6} + \dots$$

is a convergent geometric series ($a=1$, $r=1/2$).

Example 2

The alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots$$

does not converge absolutely since the series of absolute values

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots$$

diverges (it is the harmonic series).

Absolute convergence is important because of the following theorem.

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Theorem 11.7.4

If the series $\sum_{k=1}^{\infty} u_k$ converges absolutely, then the series $\sum_{k=1}^{\infty} u_k$ converges.

In other words, if a series converges absolutely, then it converges.

Example

Since the series $1 - \frac{1}{2} - \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} - \frac{1}{2^5} - \frac{1}{2^6} + \dots$ converges absolutely (Example 1), therefore by Theorem 11.7.4, the given series converges.

Conditional Convergence

If a series converges, but does not converge absolutely, then it is called conditionally convergent.

Example:

The alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots$$

converges (by Alternating Series Test), but does not converge absolutely.

Hence, it is conditionally convergent.

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Ratio Test for Absolute Convergence

Theorem 11.7.5 (Ratio Test for Absolute Convergence)

Let $\sum_{k=1}^{\infty} u_k$ be a series with nonzero terms and suppose that $\rho = \lim_{k \rightarrow \infty} \left| \frac{u_{k+1}}{u_k} \right|$ exists.

- If $\rho < 1$, the series $\sum_{k=1}^{\infty} u_k$ converges absolutely, and therefore converges.
- If $\rho > 1$ or $\rho = +\infty$, then the series $\sum_{k=1}^{\infty} u_k$ diverges.
- If $\rho = 1$, no conclusion about convergence or absolute convergence can be drawn from this test.

Example

Show that the series $\sum_{k=2}^{\infty} \frac{\cos k}{k^2}$ converges.

Solution: Since $|\cos k| \leq 1$ for all k ,

$$\left| \frac{\cos k}{k^2} \right| \leq \frac{1}{k^2}$$

The series $\sum_{k=2}^{\infty} \frac{1}{k^2}$ is a p -series with $p=2 > 1$, hence it converges.

By comparison test, $\sum_{k=2}^{\infty} \left| \frac{\cos k}{k^2} \right|$ converges.

Therefore, $\sum_{k=2}^{\infty} \frac{\cos k}{k^2}$ converges absolutely and hence converges.

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Power Series in x

If $c_0, c_1, c_2, \dots$ are constants and x is a variable, then a series of the form

$$c_0 + c_1 x + c_2 x^2 + \dots + c_k x^k + \dots = \sum_{k=0}^{\infty} c_k x^k$$

is called a power series in x .

Examples:

- $1 + x + x^2 + x^3 + \dots = \sum_{k=0}^{\infty} x^k$
- $1 - x + x^2 - x^3 + \dots = \sum_{k=0}^{\infty} (-1)^k x^k$
- $1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = \sum_{k=0}^{\infty} \frac{x^k}{k!}$

Theorem 11.8.1

For any power series in x , exactly one of the following is true:

- The series converges only for $x=0$.
- The series converges absolutely for all real x .
- The series converges absolutely for all x in some finite open interval $(-R, R)$ and diverges if $x < -R$ or $x > R$. At $x=R$ or $x=-R$, the series may converge absolutely, converge conditionally, or diverge, depending on the series.

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Radius and Interval of Convergence

From Theorem 11.8.1 (case c), the interval $(-R, R)$ is called the interval of convergence and R is called the radius of convergence.

Example

Find the interval and radius of convergence of the series

$$\sum_{k=1}^{\infty} \frac{x^k}{k}$$

Solution: Use ratio test.

$$u_k = \frac{x^k}{k}, \quad u_{k+1} = \frac{x^{k+1}}{k+1}$$

$$\rho = \lim_{k \rightarrow \infty} \left| \frac{u_{k+1}}{u_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{x^{k+1}}{k+1} \cdot \frac{k}{x^k} \right| = |x| \lim_{k \rightarrow \infty} \frac{k}{k+1} = |x|$$

Series converges if $|x| < 1$ and diverges if $|x| > 1$.

Now check the end points.

At $x=1$: $\sum_{k=1}^{\infty} \frac{1}{k}$ (harmonic series) diverges.

At $x=-1$: $\sum_{k=1}^{\infty} \frac{(-1)^k}{k}$ (alternating harmonic) converges.

Interval of convergence = $[-1, 1)$, Radius $R=1$.

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Power Series in $(x-a)$

A series of the form

$$c_0 + c_1(x-a) + c_2(x-a)^2 + \dots + c_k(x-a)^k + \dots = \sum_{k=0}^{\infty} c_k(x-a)^k$$

is called a power series in $(x-a)$.

Examples:

- $1 + (x-1) + (x-1)^2 + (x-1)^3 + \dots = \sum_{k=0}^{\infty} (x-1)^k$ ($a=1$)
- $1 - (x+3) + (x+3)^2 - (x+3)^3 + \dots = \sum_{k=0}^{\infty} (-1)^k (x+3)^k$ ($a=-3$)

Theorem 11.8.1 (for power series in $(x-a)$)

For any power series in $(x-a)$, exactly one of the following is true:

- The series converges only for $x=a$.
- The series converges absolutely for all real x .
- The series converges absolutely for all x in some finite open interval $(a-R, a+R)$ and diverges if $x < a-R$ or $x > a+R$. At $x=a-R$ or $x=a+R$, the series may converge absolutely, converge conditionally, or diverge, depending on the series.

Here the interval of convergence is centered at $x=a$ and radius of convergence is R .

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PART - 2

Lecture #45: TAYLOR AND MACLAURIN SERIES

One of the early applications of calculus was the computation of approximate numerical values for functions such as $\sin x$, $\ln x$, and e^x . One common method for obtaining such values is to approximate the function by polynomial, then use that polynomial to compute the desired numerical values.

1) PROBLEM

Given a function f and a point a on the x -axis, find a polynomial of specified degree that best approximates the function f in the "vicinity" of the point a .

Suppose that we are interested in approximating a function f in the vicinity of the point $a = 0$ by a polynomial

$$p_n(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots + c_n x^n \quad (1)$$

Because $p_n(x)$ has $n+1$ coefficients, it seems reasonable that we should be able to impose $n+1$ conditions on this polynomial to achieve a good approximation to $f(x)$. Because the point $a = 0$ is the center of interest, our strategy will be to choose the coefficients of $p_n(x)$ so that the value of p and its first n derivatives are the same as the value of f and its first n derivatives at $a = 0$. By forcing this high degree of "match" at $a = 0$, it is reasonable to hope that $f(x)$ and $p_n(x)$ will remain close over some interval (possibly quite small) centered at $a = 0$. Thus, we shall assume that f can be differentiated n times at 0, and we shall try to find the coefficients in (1) such that

$$f(0) = p(0), \quad f'(0) = p'(0), \quad f''(0) = p''(0), \quad \dots, \quad f^{(n)}(0) = p^{(n)}(0) \quad (2)$$

We have

$$\begin{aligned} p_n(x) &= c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots + c_n x^n \\ p'_n(x) &= c_1 + 2c_2 x + 3c_3 x^2 + \dots + n c_n x^{n-1} \\ p''_n(x) &= 2c_2 + 3 \cdot 2 c_3 x + \dots + n(n-1) c_n x^{n-2} \\ p'''_n(x) &= 3 \cdot 2 c_3 + 4 \cdot 3 \cdot 2 c_4 x + \dots + n(n-1)(n-2) c_n x^{n-3} \\ &\vdots \\ p^{(n)}_n(x) &= n(n-1)(n-2) \dots 2 \cdot 1 c_n = n! c_n \end{aligned}$$

In general,
 $p_n^{(k)}(x) = k! c_k +$
 terms containing x
 (which vanish at $x = 0$)

Thus, to satisfy (2) we must have

$$\begin{aligned} f(0) &= p(0) = c_0 \\ f'(0) &= p'(0) = c_1 \\ f''(0) &= p''(0) = 2c_2 \\ f'''(0) &= p'''(0) = 3 \cdot 2 c_3 \\ &\vdots \\ f^{(n)}(0) &= p^{(n)}(0) = n(n-1)(n-2) \dots 2 \cdot 1 c_n = n! c_n \end{aligned}$$

Hence;

$$\begin{aligned} c_0 &= f(0) \\ c_1 &= f'(0) \\ c_2 &= \frac{f''(0)}{2!} \\ c_3 &= \frac{f'''(0)}{3!} \\ &\vdots \\ c_n &= \frac{f^{(n)}(0)}{n!} \end{aligned}$$

These are
 the values of
 coefficients of
 $p_n(x)$.

Therefore;

$$c_0 = f(0), \quad c_1 = f'(0), \quad c_2 = \frac{f''(0)}{2!}, \quad c_3 = \frac{f'''(0)}{3!}, \quad \dots, \quad c_n = \frac{f^{(n)}(0)}{n!}$$

Lecture #45: TAYLOR AND MACLAURIN SERIES

Part - 2

Example: Find the Maclaurin polynomials P_0, P_1, P_2, P_3 and P_n for e^x .

Solution: Let $f(x) = e^x$. Thus

$$f^{(0)}(x) = e^x, f'(x) = e^x, f''(x) = e^x, f^{(3)}(x) = e^x, \dots, f^{(n)}(x) = e^x$$

So,

$$f(0) = e^0 = 1, f'(0) = 1, f''(0) = 1, f^{(3)}(0) = 1, \dots, f^{(n)}(0) = 1$$

$$f^{(n)}(0) = 1 \text{ for all } n \geq 0$$

From the coefficients formula (from Part-1):

$$c_n = \frac{f^{(n)}(0)}{n!} = \frac{1}{n!}$$

Now, the Maclaurin polynomials are

$$P_0(x) = f(0) = 1$$

$$P_1(x) = f(0) + f'(0)x = 1 + x$$

$$P_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 = 1 + x + \frac{x^2}{2!}$$

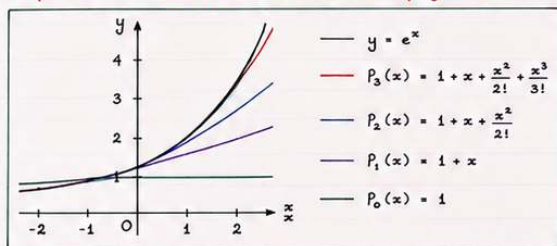
$$P_3(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3$$

$$= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$$

$$\vdots$$

$$P_n(x) = \sum_{k=0}^n \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$$

Graphs of e^x and first four Maclaurin polynomials



Note:

- ★ Near the origin, $P_1(x), P_2(x), P_3(x)$ are almost indistinguishable from e^x .
- ★ Away from the origin, the approximation is not good.

To obtain polynomial approximations of $f(x)$ that have their best accuracy near a general point $x = a$, it will be convenient to express polynomials in powers of $x - a$, so that they have the form

$$P(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots + c_n(x-a)^n$$

TAYLOR AND MACLAURIN SERIES

For a fixed value of x near a , one would expect that the approximation of $f(x)$ by its Taylor polynomial $P_n(x)$ about $x = a$ should improve as n increases,

⇒ Increasing n matches higher and higher derivatives of $f(x)$ with those of $P_n(x)$ at $x = a$.

⇒ We might be able to achieve any desired degree of accuracy by choosing n sufficiently large; that is, $P_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$.

In the special case where $a = 0$, the Taylor series for f is called the **Maclaurin series** for f .

★ TAYLOR POLYNOMIAL OF DEGREE n ABOUT $x = a$

If f is n times differentiable at $x = a$, then the Taylor polynomial of degree n of f about $x = a$ is

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$$

where $f^{(k)}(a)$ denotes the k^{th} derivative of f at $x = a$.

It matches $f(x)$ and its first n derivatives at $x = a$.

★ MACLAURIN SERIES ($a = 0$)

The Maclaurin polynomial of degree n is

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} x^k$$

Hence the Maclaurin series of $f(x)$ is

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k$$

IMPORTANT NOTES

- ★ $P_n(x)$ is a polynomial of degree n .
- ★ Maclaurin series is simply Taylor series about $a = 0$.
- ★ These series are used for approximating functions near the point a .
- ★ More terms in the series ⇒ better approximation near a .
- ★ But the approximation may not be good far away from a .